

# WHERE THE SECONDS GO: BOUNDING MIXTURE-OF-EXPERTS DECODE FROM HOST MEMORY, AND MEASURING WHAT FORESIGHT IS WORTH

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## ABSTRACT

A Mixture-of-Experts model too large for one consumer GPU keeps most of its experts in host memory, and decoding one token at a time then waits on host reads. We bound its time per token two ways: by the fewest host reads any policy can make with  $C$  GPU slots per layer (Belady’s MIN with bypass) over the machine’s measured read rate, and, for systems that read an expert only once it is requested, by the same reads in series with the GPU’s own compute. Across 30 launches on 21 rented RTX 5090 machines, our llama.cpp expert cache, faster than FreeToken at 11 of 12 configurations, takes the GPU’s profiled compute plus its host reads at the machine’s best rate, with nothing fitted (within 6% on 28 of 30 launches of gpt-oss-120b at the smallest budget; over-predicted at the next, where background copies overlap the GPU’s work): its gap to the first bound is that serialisation (22–51% at the smallest budget) plus reads MIN would not make. Oracles of the future routing built into the engine show that foresight pays when spent as MIN spends it, caching MIN’s set and reading each expert once, and that which of MIN’s many hit-optimal sets is used matters: the greedy one loses on machines with a slow link, where the set with the fewest admissions gains. Degrading the oracle prices foresight: half of MIN’s saving over admitting every miss needs the routing of about 0.65  $C$  distinct experts per layer ahead, which no online policy, learned admission order, speculative batch or forecaster we tested approaches.

## 1 INTRODUCTION

Open-weight Mixture-of-Experts (MoE) models now carry 30–120 B parameters and activate a few billion per token; a desktop GPU has 24–32 GB. Running such a model for one user therefore means keeping most experts in host DRAM. Each decoded token reads its routed experts from GPU memory if they are resident, and otherwise from host memory: either the CPU runs the expert, or the expert is copied over PCIe into a GPU slot (Ivanov et al., 2021; Hoefler et al., 2021). Many systems build on this: GPU expert caches with demand fetch (Eliseev & Mazur, 2023; Xue et al., 2024; Song et al., 2024a), CPU execution of misses (Kamahori et al., 2024; Chen et al., 2025a; Zhong et al., 2025; Huang et al., 2025; Zhang et al., 2025; Wang et al., 2026a), a per-miss choice between the two (Yang et al., 2026; Zhu et al., 2026), and expert prediction (Hwang et al., 2024; Zhu et al., 2025; Wang et al., 2026b). Each reports a speed-up over a baseline of its choosing. None states how fast the same machine *could* decode the model within the same GPU memory, or what the rest of the time is made of.

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We answer both questions for batch-1 decode of gpt-oss-120b (OpenAI, 2025) and Qwen3-30B-A3B (Yang et al., 2025) on rented RTX 5090 hosts, with three findings.

**(1) Where host memory binds, time is GPU compute plus host bytes** (Sections 3 and 4). The fewest host reads any exact-routing policy can make, over the machine’s measured read rate, bounds the time per token; for a system that reads on demand, the GPU’s own compute adds in series. Our cache’s time has the same series form with its own reads: for gpt-oss-120b at 11%, the profiled GPU compute (median 4.3 ms) plus its counted reads at the machine’s best rate is within 6% of the measured time on 28 of 30 launches on 21 machines, with nothing fitted, and on all 4 launches where we registered the relation before the runs. The first set is where we found the relation; the registered launches test it. At 25% it over-predicts (median 6%) and failed its registered test, which we attribute to background copies overlapping the GPU’s work. At 11% its gap to the first bound is therefore that serialisation (22–51% of the gap) plus the reads it makes beyond MIN’s. The bounds’ read time is nearly reachable: MIN’s reads, replayed layer by layer without the model, reach 86–96% of it, and requiring every admission to cross the link raises it by at most 7%.

**(2) Foresight pays when spent as MIN spends it, and which MIN matters** (Section 4). With the future routing known, caching what MIN caches and copying each expert

once, in the step, is 16–51% faster than our deployed cache on three hosts at six budgets. MIN has many hit-optimal schedules: the greedy one admits so eagerly that on the 2 machines whose link reads at under a third of their CPU rate it loses ( $0.86\text{--}0.91\times$ ), while the one with the fewest admissions gains there ( $1.03\text{--}1.04\times$ ); on the 5 machines whose link reads at  $0.28\text{--}0.63$  of their CPU rate it adds  $0.08\text{--}0.17$  to the speed ratio at gpt-oss 11%. An order-free (Shapley) accounting shows that what to cache and how to load interact: either alone closes at most 26% of the gap, both together up to 60%. A model of the two read paths predicts the deployed cache and the in-step states within a median 2.2% on 14 hosts.

**(3) Foresight has a measurable price** (Section 5). Degraded to a window of the next  $W$  tokens, the oracle gains little over the deployed cache: a 16-token window on the deployed path at most 12%, and shorter or half-right windows  $0.94\text{--}1.03\times$ . On nine models’ traces, the window that closes half of the read gap holds about  $0.65C$  distinct experts per layer, a one-constant rule that, given a model’s routing trace, predicts its horizon as well as a two-constant power law. No online policy, learned admission order, speculative batch or forecaster we tested comes near it.

## Contributions.

- **Two bounds on time per token** for exact-routing systems with  $C$  slots per layer, from MIN’s host reads and the machine’s probed rates: one for systems with foresight, one for systems that read on demand, with a microbenchmark and a link-aware variant showing their read time is nearly reachable (Section 3).
- **A relation with nothing fitted:** for gpt-oss-120b at 11%, time per token is the GPU’s profiled compute plus the counted host reads at the machine’s best rate, within 6% on 28 of 30 launches and on all 4 launches where it was registered before the runs; it splits the gap into serialisation and excess reads, and over-predicts at 25%, where background copies overlap the GPU’s work (Section 4).
- **An in-engine oracle factorial** that crosses what to cache with how each expert is loaded, on 15 hosts, with an order-free accounting, a test of launch-to-launch variation, a model of the two read paths tested on hosts it was not fitted on, and a comparison of MIN’s greedy and fewest-admission schedules in the engine (Section 4).
- **A price on foresight:** its value by horizon and accuracy, in the engine and on nine models’ traces, and a one-constant rule for the horizon in distinct experts (Section 5).
- An artifact in which every prediction was committed

before the machine it concerns started, and every number is generated by a script (Appendices B and L).

## 2 SETTING AND METHOD

**Models and workload.** gpt-oss-120b (36 MoE layers of 128 experts, top-4, MXFP4, 13.25 MB per expert) and Qwen3-30B-A3B in BF16 (48 layers of 128 experts, top-8, 9.44 MB per expert), on the 30 AIME-25 problems; the cache carries over between problems, as in one user’s session. The system comparison decodes 256 tokens per problem greedily through each system’s server; the oracle jobs replay each model’s own greedy output teacher-forced, so that every configuration sees the same routing. A *budget* is the share  $C/E$  of each layer’s experts that fit in GPU slots: 11%, 25% and 40% for gpt-oss, 12.5%, 25% and 43.75% for Qwen3.

**The engine.** Our cache is a 3,500-line patch to llama.cpp (Gerganov & contributors, 2023–2026). Each MoE layer has  $C$  GPU slots. Routing is decided on the GPU; resident experts run there. A *miss* is read from host memory in one of two ways: CPU helper threads run it, or it is *fetched*, copied into a slot and run on the GPU in the same step. The deployed policy admits a miss when its decayed request count (half-life 16 tokens) beats the lowest resident’s by a margin  $\kappa=1$ ; the CPU serves the admitted miss and a background copy then fills its slot, so it is read from host memory twice. How many of a layer’s misses to fetch rather than run on the CPU is a per-machine table computed from the machine’s bandwidth probe (Appendix C).

**Configurations and terms.** Table 1 names the policies. The oracles read the routing of the rest of the problem from a trace recorded on the same machine; they decide, they do not predict. *MIN* is Belady’s optimum with bypass (Belady, 1966; Zhang et al., 2020): it *admits* a miss into a slot only if some resident is needed later than the miss, and otherwise *bypasses* it, serving it without admission; with bypass, this maximises hits under  $C$  slots (Carlisle & Lloyd, 1995; Jain & Lin, 2016). An admitted expert is *loaded* in one of two ways: *in the step*, copied over PCIe before the GPU runs it (one host read, on the critical path), or *by the CPU*, served there and copied in the background (two reads). The two *read paths* are the CPU’s own reads of host memory and the PCIe link. A budget is *host-bound* when host reads, not the GPU, bound the time. The *gap* is the deployed cache’s time per token minus the bound’s.

**Hosts and statistics.** RTX 5090 cards (32 GB, 1.79 TB/s datasheet) in desktop hosts, rented by the hour. Every job first probes how fast the host reads its memory: from CPU threads ( $B_c$ ), over PCIe ( $B_p$ ), both at once ( $B_{cp}$ ), and the highest rate by any method ( $B_{\text{host}}$ );  $B_p/B_c$  is the host’s *link-*

Table 1. Cache configurations. *Reads*: how many times an admitted expert is read from host memory. All run in the same engine; the oracles read the future routing from a trace of the same text recorded on the same machine.

| Name                    | What it admits, and how                                                                                                                                         | Reads  |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <i>No foresight</i>     |                                                                                                                                                                 |        |
| deployed                | decayed request count beats the lowest resident’s by $\kappa=1$ ; the CPU serves the miss, then a background copy fills the slot                                | 2      |
| single read             | the same admissions, fetched into the slot in the step                                                                                                          | 1      |
| admit every miss        | every miss fetched into the lowest-scored slot (demand fetch)                                                                                                   | 1      |
| <i>Oracles</i>          |                                                                                                                                                                 |        |
| MIN, 2 reads            | MIN with bypass (greedy rule); the CPU serves the miss, then a copy                                                                                             | 2      |
| MIN, 1 read             | MIN with bypass (greedy rule); admissions fetched in the step                                                                                                   | 1      |
| MIN, fewest adm.        | the hit-optimal schedule with the fewest admissions (an LP per host), with 2 or 1 reads                                                                         | 2 or 1 |
| MIN, prefetched         | MIN, 1 read, plus copies issued three steps ahead                                                                                                               | 1      |
| Belady                  | every expert of the coming steps, into the slot needed furthest ahead (also with 1 read, and prefetched)                                                        | 2 or 1 |
| window $W$              | admit every miss unless the next $W$ tokens show the victim needed sooner; victim the lowest-scored resident not seen in the window, else the one seen furthest | 1      |
| window $W$ , recall $r$ | the same, each future expert kept with probability $r$ , else replaced by a random one                                                                          | 1      |
| deployed + window $W$   | the deployed policy with the same window and victim choice; the CPU serves the miss, then a background copy                                                     | 2      |

*to-CPU ratio*. Time per token is averaged over problems, ratios are ratios of means paired by problem, and within-host intervals are 95% bootstrap percentiles over problems. Machines differ far more than problems or launches: two panel hosts with the same CPU model differ by 16–29% in the deployed cache’s time, while 4 machines relaunched later came within 1.9% of their first launch and every ratio within 0.030, more than a typical within-host interval, so we do not read smaller differences on one host. A host panel (10 hosts, the first 20 problems, configurations in a seeded random order) therefore treats the host as the unit: pooled intervals come from a crossed bootstrap over hosts and problems (Kalibera & Jones, 2013). Each job’s predictions were committed before its machine started; Appendix B scores those of jobs 073 onward.

### 3 A BOUND ON TIME PER TOKEN

Consider any system that executes the model’s exact routing with at most  $C$  experts of each layer resident on the GPU. Per token each of the  $Lk$  routed experts is either served by a resident copy, which the GPU reads from its own memory, or read from host memory: by the CPU that runs it, or by the copy that brings it into a slot. Let  $R^*$  be MIN’s host reads per token, summed over layers, on the routing trace; no policy, even one that knows the future, reads fewer. If  $c$  experts per token run on the CPU, host memory supplies at least  $\max(R^*, c)S$  bytes of experts of  $S$  bytes each, and the GPU reads the dense weights  $D$  plus  $(Lk - c)S$ . With everything overlapped,

$$\bar{T} \geq \min_{0 \leq c \leq Lk} \max \left( \frac{D + (Lk - c)S}{B_{\text{gpu}}}, \frac{\max(R^*, c)S}{B_{\text{host}}} \right). \quad (1)$$

A budget is *host-bound* when the minimum is at  $c=0$ : the GPU could take every expert, and host reads bind. That holds at the two smallest budgets of each model. The bound covers exact routing, whole experts of unmodified bytes, at most  $C$  residents per layer and one token per forward pass; it drops all latencies and lets the two read paths share  $B_{\text{host}}$  freely. Its MIN runs over the problems in sequence and may evict an expert in the step that served it; the oracles and replays of Sections 4 and 5 restart MIN at each problem and keep each step’s experts, which reads a few percent more.

#### A tighter bound for systems that read on demand.

Equation (1) overlaps the GPU’s work with the host reads. At batch size 1 a system that reads an expert only after its layer’s router has chosen it cannot: layer  $l$ ’s misses are known only after layer  $l$ ’s attention and router, and layer  $l+1$  starts only after layer  $l$ ’s experts. Its time is therefore at least the GPU’s non-expert work plus the reads, in series:

$$\bar{T}_{\text{demand}} \geq T_{\text{GPU}} + \frac{R^*S}{B_{\text{host}}}, \quad (2)$$

where  $T_{\text{GPU}}$ , the GPU’s attention, dense layers, norms and routers per token, is 2.9–3.4 ms for gpt-oss in Nsight profiles of our engine on 7 hosts. Every system we compare, ours included, reads on demand. Only reading before routing, that is foresight, can approach Eq. (1); Eq. (2) is the bound for everything else.

Table 2 places three systems against it on two hosts. Our cache leads FreeToken (Yang et al., 2026) at 11 of 12 configurations and runs 2.0–4.0 $\times$  stock llama.cpp at equal GPU expert memory. Tuned over five settings on a third host, FreeToken gains at most 1.04 $\times$  and still trails at 5 of six cells (Appendix J). Against Eq. (1), our cache stands at 25–43%; with the GPU term at the rate all-in-VRAM decode reaches at batch size 1, at 31–56%. On the 21 machines of Section 4 at gpt-oss 11%, it stands at 38–54% of Eq. (1) and at 55–70% of Eq. (2).

Table 2. Decode speed at equal GPU expert memory on two RTX 5090 hosts (30 AIME-25 problems, the first 256 greedy tokens, one session). Ratios are of mean speeds, paired by problem, with 95% bootstrap intervals. Our cache uses each machine’s fetch table; FreeToken uses its faster backend per budget, picked on host B on a separate launch. *Bound*: Eq. (1) on that machine (MIN’s reads, the host’s highest probed rate) with the GPU at its datasheet rate, and at the rate all-in-VRAM decode measures at batch size 1 (52% of datasheet for gpt-oss, 61% for Qwen3; Table 18 varies the other assumptions). Against the pooled bound their designs allow, FreeToken (one cache for all layers) and llama.cpp (whole layers pinned) stand at 18–40% and 8–20%. With every weight in GPU memory (an RTX PRO 6000) stock llama.cpp decodes gpt-oss at 255 and Qwen3 at 178 tok/s. <sup>†</sup>Selection launch only, which favours FreeToken.

| Model                                                                                                                                                           | Experts on GPU | llama.cpp (tok/s) | FreeToken (tok/s)     | Ours (tok/s) | Ours ÷ FreeToken (95% CI)         | Ours ÷ llama.cpp | Bound, datasheet GPU tok/s | Bound, measured GPU tok/s | GPU ours |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|-----------------------|--------------|-----------------------------------|------------------|----------------------------|---------------------------|----------|
| <b>Host B:</b> Ryzen 9 9950X3D, card at 17001 MHz memory clock, CPU 72 GB/s, link 53, together 78, highest (the bound’s rate) 88                                |                |                   |                       |              |                                   |                  |                            |                           |          |
| gpt-oss-120b                                                                                                                                                    | 11%            | 34.9              | 54.0 (hybrid)         | <b>69.9</b>  | 1.294 [1.278, 1.312]              | 2.00×            | 172                        | 41%                       | 172      |
| gpt-oss-120b                                                                                                                                                    | 25%            | 40.0              | 85.6 (hybrid)         | <b>109.2</b> | 1.275 [1.254, 1.295]              | 2.73×            | 430                        | 25%                       | 279      |
| gpt-oss-120b                                                                                                                                                    | 40%            | 47.7              | 132.2 (offload)       | <b>152.5</b> | 1.154 [1.134, 1.172]              | 3.20×            | 519                        | 29%                       | 279      |
| Qwen3-30B-A3B                                                                                                                                                   | 12.5%          | 19.5              | 38.7 (hybrid)         | <b>40.0</b>  | 1.032 [1.022, 1.042]              | 2.05×            | 96                         | 42%                       | 96       |
| Qwen3-30B-A3B                                                                                                                                                   | 25%            | 22.1              | 54.7 (hybrid)         | <b>63.1</b>  | 1.153 [1.134, 1.173] <sup>†</sup> | 2.86×            | 214                        | 30%                       | 192      |
| Qwen3-30B-A3B                                                                                                                                                   | 43.75%         | 28.4              | 103.1 (offload)       | <b>108.2</b> | 1.049 [1.036, 1.064]              | 3.81×            | 308                        | 35%                       | 192      |
| <b>Host S:</b> Ryzen 9 9950X at the stock clock, card at 14001 MHz, CPU 56 GB/s, link 53, together 66, highest 71; FreeToken’s backend carried over from host B |                |                   |                       |              |                                   |                  |                            |                           |          |
| gpt-oss-120b                                                                                                                                                    | 11%            | 28.9              | 48.0 (hybrid)         | <b>57.9</b>  | 1.207 [1.195, 1.219]              | 2.00×            | 140                        | 41%                       | 140      |
| gpt-oss-120b                                                                                                                                                    | 25%            | 33.6              | 78.7 (hybrid)         | <b>94.1</b>  | 1.196 [1.163, 1.224]              | 2.80×            | 351                        | 27%                       | 275      |
| gpt-oss-120b                                                                                                                                                    | 40%            | 40.1              | 122.2 (offload)       | <b>133.6</b> | 1.093 [1.077, 1.111]              | 3.34×            | 514                        | 26%                       | 275      |
| Qwen3-30B-A3B                                                                                                                                                   | 12.5%          | 16.1              | 32.7 (hybrid)         | <b>33.5</b>  | 1.027 [1.018, 1.036]              | 2.08×            | 78                         | 43%                       | 78       |
| Qwen3-30B-A3B                                                                                                                                                   | 25%            | 18.5              | 51.4 (hybrid)         | <b>53.9</b>  | 1.048 [1.037, 1.059]              | 2.91×            | 174                        | 31%                       | 174      |
| Qwen3-30B-A3B                                                                                                                                                   | 43.75%         | 23.8              | <b>98.3</b> (offload) | 95.8         | 0.974 [0.962, 0.987]              | 4.02×            | 305                        | 31%                       | 189      |

**How tight are they?** The GPU term of Eq. (1) is loose: at batch size 1 all-in-VRAM decode reaches only 52% (gpt-oss) and 61% (Qwen3) of the datasheet rate, which moves the bound at the larger budgets but not where the host term binds. With every assumption of Table 18 relaxed at once, our cache stands at 33–57% of the bound on host B. The host term is nearly reachable. A microbenchmark (job 102) replays MIN’s per-layer reads of gpt-oss at 11% and 25% on three hosts without the model, CPU threads reading part of each layer’s experts while the copy engine copies the rest into slots: waiting for every layer, it reaches 86–96% of the host term (Table 19). It runs no compute, and lets every read take either path; but an expert that is admitted and read once must cross the link. MIN has many hit-optimal schedules, and on the 30-problem trace the fewest admissions any of them needs is 12.4 per token at gpt-oss 11%, against 21.8 for the greedy rule (a linear program over reuse intervals, `scripts/linkaware.py`). Sending every admission over the link at the probe’s highest link rate raises the bound by at most 1% at 11% and 7% at 25% on the 24 probes of jobs 095–102. Pooling the *LC* slots across layers would lower the reads by 4.5–18.6%; we score FreeToken’s single cache against that pooled bound too (18–40%).

**Published systems.** Scored by one procedure (the same bound at each paper’s datasheet ceilings, MIN’s reads from our traces of its model; Appendix K), the 20 published batch-1 measurements inside the bound’s class (lossless, non-speculative, with a routing trace of their model) reach a median 13.6% of it (quartiles 8.1–20.6%), and our cache 18–42%. Systems in this class run several times slower than the routing allows on the same machine; the next section asks what the missing time is made of.

## 4 WHERE THE SECONDS GO

The deployed cache does not know which experts the coming tokens will need. We first ask what its time is made of, then measure what knowing the future is worth by running oracles inside the engine: on three hosts (O3–O5) at all six budgets, on the 10-host panel at the two host-bound gpt-oss budgets, and on follow-up launches, 34 launches on 23 machines in all. A running example: on O4 (a Ryzen 9 9950X) at gpt-oss 11%, Eq. (1) gives 10.0 ms per token, Eq. (2) 13.0, and the deployed cache takes 20.6; MIN copying each admission in the step takes 15.2, and with the copies made a few steps ahead, 13.8.

**Time is GPU compute plus host reads.** For the deployed cache at gpt-oss 11%, one relation with nothing fitted describes the launches:

$$T = G + \frac{RS}{B_{\text{host}}}, \quad (3)$$

with  $R$  the host reads per token the engine counts (misses and admissions) and  $G$  the GPU’s own kernel time per token, 4.1–4.7 ms in 14 Nsight profiles on 7 hosts. With  $G$  the median of the profiles at this budget (4.3 ms), Eq. (3) is within 6% of the measured time on 28 of the 30 launches (19 of 21 machines; median absolute error 1.1%, range –10 to 14%). These launches are where we found the relation, and their counted reads span only 63–69 per token, so they test the host rate more than the read count. Solved for  $G$  on each launch, it gives a median of 4.3 ms (interquartile range 4.1–4.6), uncorrelated with the link-to-CPU ratio (Spearman over launches 0.02). With  $G$  removed, the time scales with the host’s read rate with an elasticity of –0.95 (95% interval over machines –1.06 to –0.84), as pure reads at

$B_{\text{host}}$  would. We then registered Eq. (3) before job 105’s runs, with  $G$  profiled on each host before its timed runs (4.1–4.5 ms) and  $R$  from each run’s counters: at 11% it came within  $-1.5$  to  $4.7\%$  on all 4 launches, two of them on new machines (Table 4). At 25% the relation over-predicts: by a median 6% on the earlier launches (17 of 30 within 6%), and in job 105 by 6.9–9.3% on 3 of the 4 hosts (3.8% on the fourth), beyond the 6% we had registered; the pooled median error, 4.2%, also exceeded the 4% registered. We attribute this to the background admissions, a larger share of the reads there, overlapping the GPU’s work (the implied  $G$  is 3.8 ms), but have not measured that overlap directly. The deployed cache thus executes its own reads at the machine’s best rate, and at 11% its whole gap to Eq. (1) is two terms: the GPU’s compute in series with the reads, 22–51% of the gap with each launch’s implied  $G$  (31–55% at 25%), and the reads it makes beyond MIN’s. Foresight can attack both. Fewer reads need MIN’s choices; overlap needs reads issued ahead, and with the copies made ahead the implied  $G$  falls to 2.7 ms. Copies in the step do the opposite: they add latency on the critical path that grows as the link slows (implied  $G$  7.4 ms for MIN, correlation  $-0.91$  with the ratio; 5.5 ms for the deployed set; 16.4 ms when admitting every miss), so Eq. (3) with the profiled  $G$  describes the deployed cache only. The calibrated model of Appendix C extends Eq. (3) to two read paths,  $T = G' + \max(X_c/B_c, X_p/B_p, (X_c + X_p)/B_{cp})$  with  $X_c$  and  $X_p$  the bytes the CPU and the link read and  $G'$  one constant per budget fitted on other hosts, and predicts the deployed cache and the in-step states within a median 2.2% on 14 hosts; it is optimistic for copies in the step on slow links and for copies in the background.

**Spent as MIN spends it, foresight pays.** MIN with each admission copied once, in the step, is faster than the deployed cache at every budget on O3–O5, by 16–51%, and by 25–81% with the copies made ahead. The usual ways of spending foresight waste reads: Belady prefetch admits experts MIN would bypass, and serving a miss on the CPU before copying it reads every admission twice. They read  $1.6\text{--}2.5\times$  MIN’s bytes and, at the smallest budget of each model on O4 and O5, gain at most 15% or lose, where MIN with one read gains  $1.16\text{--}1.44\times$ ; the trade-off is classical for disks (Cao et al., 1995) and CPU caches (Jain & Lin, 2018). Even MIN with its copies made ahead reaches only 46–73% of the bound on O4 and O5 at the host-bound budgets.

**Which of MIN’s schedules matters.** MIN has many hit-optimal schedules. The greedy rule our oracles follow admits a miss whenever some resident is needed later, and may evict it again before its next use; the schedule with the fewest admissions among those with the most hits (Section 3) admits far fewer. Jobs 104 and 105 gave the engine that schedule, computed on each host from its own rout-

Table 3. Which of MIN’s hit-optimal sets the oracle follows (job 104): speed relative to the deployed cache on the same host, with MIN’s set loaded by the CPU (two reads) or in the step (one read), as the greedy rule picks it or as the schedule with the fewest admissions does; and the copies per token each makes in the step. Hosts sorted by the probe’s link-to-CPU ratio.

| Host             | Link/<br>CPU | Budget      | By the CPU |        | In the step |        | Copies/token |        |
|------------------|--------------|-------------|------------|--------|-------------|--------|--------------|--------|
|                  |              |             | greedy     | fewest | greedy      | fewest | greedy       | fewest |
| Pf again         | 0.29         | gpt-oss 11% | 0.94       | 1.11   | 0.86        | 1.04   | 19.8         | 11.8   |
| Pf again         | 0.29         | gpt-oss 25% | 1.13       | 1.22   | 0.96        | 1.06   | 9.7          | 7.0    |
| 285K, second     | 0.49         | gpt-oss 11% | 1.03       | 1.08   | 1.05        | 1.18   | 19.8         | 11.8   |
| 285K, second     | 0.49         | gpt-oss 25% | 1.20       | 1.21   | 1.12        | 1.17   | 9.7          | 7.0    |
| Pg again (5950X) | 0.63         | gpt-oss 11% | 1.02       | 1.06   | 1.26        | 1.34   | 19.8         | 11.8   |
| Pg again (5950X) | 0.63         | gpt-oss 25% | 1.20       | 1.15   | 1.32        | 1.35   | 9.7          | 7.0    |

Table 4. Job 105, registered before the runs, nothing fitted: the deployed cache’s time per token against Eq. (3), the GPU’s compute profiled on the same host before the timed runs plus the run’s counted host reads at the machine’s best rate; and the speed, relative to the deployed cache, of the deployed cache with the engine’s layer-ahead copy (each layer’s top predicted expert of the next layer copied ahead of use) and of MIN’s greedy and fewest-admission sets copied in the step. Hosts sorted by the probe’s link-to-CPU ratio.

| Host               | Link/<br>CPU | Budget      | $G_{\text{prof}}$<br>(ms) | Deployed (ms) |          | Layer-<br>ahead | MIN, in step |        |
|--------------------|--------------|-------------|---------------------------|---------------|----------|-----------------|--------------|--------|
|                    |              |             |                           | law           | measured |                 | greedy       | fewest |
| Pf again           | 0.28         | gpt-oss 11% | 4.1                       | 13.6          | 13.9     | 0.72            | 0.86         | 1.03   |
| Pf again           | 0.28         | gpt-oss 25% | 4.4                       | 9.8           | 9.5      | 0.69            | 0.97         | 1.06   |
| Threadripper 9960X | 0.32         | gpt-oss 11% | 4.1                       | 9.3           | 9.1      | 0.91            | 0.91         | 1.04   |
| Threadripper 9960X | 0.32         | gpt-oss 25% | 4.4                       | 7.3           | 6.8      | 0.87            | 0.98         | 1.04   |
| EPYC 7402P         | 0.45         | gpt-oss 11% | 4.3                       | 19.1          | 18.2     | 0.81            | 1.01         | 1.15   |
| EPYC 7402P         | 0.45         | gpt-oss 25% | 4.5                       | 12.9          | 11.8     | 0.78            | 1.09         | 1.15   |
| O4 again (9950X)   | 1.04         | gpt-oss 11% | 4.4                       | 20.8          | 20.5     | 1.04            | 1.36         | 1.34   |
| O4 again (9950X)   | 1.04         | gpt-oss 25% | 4.4                       | 13.2          | 12.4     | 1.01            | 1.42         | 1.39   |

ing trace, on 7 launches on 6 machines whose link reads at 0.28–1.04 of their CPU rate (Tables 3 and 4). On the engine’s 20 problems it copies 0.60–0.73 times as many experts in the step as the greedy schedule, for 2.5–5.3% more misses. On the 5 machines whose link reads at 0.28–0.63 of their CPU rate it adds 0.08–0.17 to the greedy schedule’s speed ratio at gpt-oss 11% (6 launches); on the machine whose link is as fast as its CPU, where copies in the step are cheap, it changes it by  $-0.01$ . On the 2 machines whose link reads at under a third of their CPU rate, the greedy schedule loses ( $0.86\text{--}0.91\times$ ) and the fewest-admission one gains ( $1.03\text{--}1.04\times$ , 3 launches). On every launch the fewest-admission set gains over the deployed cache with either load and at both budgets, each 95% interval above 1; how much depends on the link, and loaded by the CPU on the machine with the fastest link it is 0.07 behind the greedy set at 25%.

**The machine decides how much.** Across machines, the gain of every state that moves reads from the CPU to the link follows the link-to-CPU ratio (Spearman over the machines that ran it through job 104, 95% bootstrap interval): MIN copied in the step 0.89 [0.63, 0.97] (19 machines), copied ahead 0.87 [0.54, 0.99] (17), admitting every miss 0.95 [0.75, 0.99] (13; Fig. 2). MIN loaded by the CPU, which

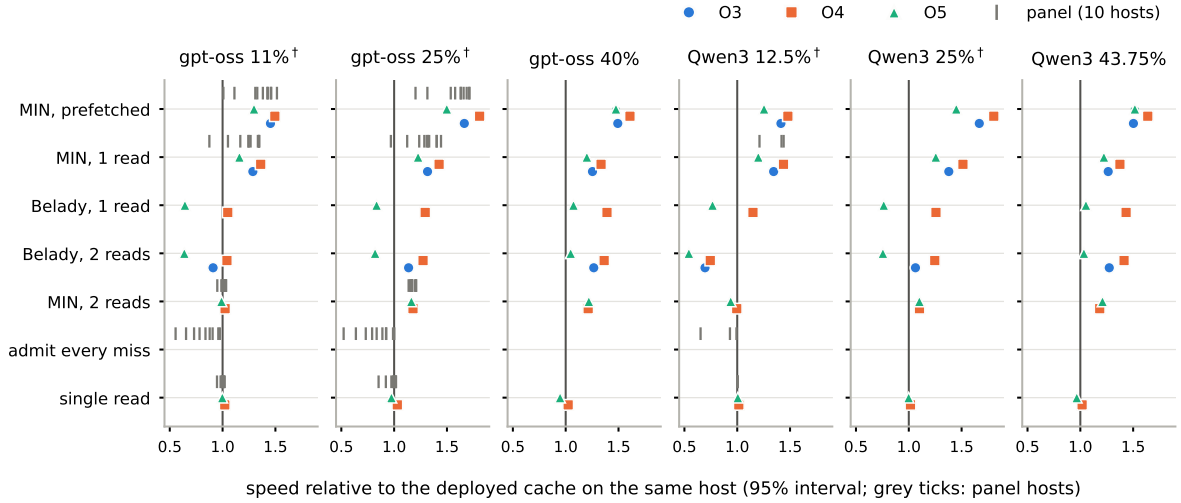


Figure 1. What foresight is worth in the engine, by how it is spent: speed of each configuration of Table 1 relative to the deployed cache on the same host, at the six budgets (<sup>†</sup>host-bound). Hosts O3, O4 and O5 at every budget (O4 and O5 in a randomised order); the 10 panel hosts (grey ticks) at the two gpt-oss host-bound budgets, and 3 of them at Qwen3 12.5%. The prefetched Belady variant and the learned admission order are in Appendices G and H. Within-host 95% intervals are mostly narrower than the markers. On these hosts admitting what MIN admits and reading it once (*MIN, 1 read*) gains at every budget; the usual ways of spending foresight (*Belady, MIN, 2 reads*) gain at some budgets and lose at others. On the 2 machines whose link reads at under a third of their CPU rate, MIN’s greedy schedule loses and its fewest-admission one gains (Fig. 2 and Tables 3 and 4).

reads every admission twice and copies it in the background, tracks the link’s absolute rate more closely (0.79 [0.49, 0.92], 17 machines, against 0.45 for the ratio; the two intervals overlap, and the feature was picked from a search over all probe features), which suggests that a background copy needs a fast link and an in-step copy a balanced one. The greedy schedule copied in the step still gains  $1.02\text{--}1.05\times$  at 11% on 2 further machines with ratios 0.40 and 0.48, and loses only on the 2 machines whose link reads at under a third of their CPU rate; made ahead of use, the same copies lose beyond their interval on no machine they ran on (at worst  $0.997\times$ , on Pf).

**Machines differ more than problems.** Problems rank largely the same on every machine: Kendall’s coefficient of concordance of MIN-copied-in-the-step’s per-problem gain over 19 machines and 20 problems is 0.77 at 11% and 0.82 at 25%, and a problem’s gain follows how far the deployed policy’s reads exceed MIN’s on its trace (Spearman 0.88 at 11%, 0.68 at 25%). Yet machines carry 96% of the variance of its log gain and problems 1–3%, so the statements of this section are about orderings that hold on every host of a stated class, and about a model that predicts each host. An order-free (Shapley) accounting of what to cache against how to load is in Appendix E: the two choices interact, and together close up to 60% of the gap to Eq. (1), either alone at most 26%.

## 5 HOW MUCH FORESIGHT IS NEEDED

The oracles know the whole future; a forecaster would know part of it, imperfectly. We give the cache a *window*, the routing of the next  $W$  tokens, exact or with errors, and ask what it is worth in the engine and, on the traces of nine models, how long it must be.

**Window policies.** *Window*  $W$  extends admitting every miss: at each step, each miss in request order takes a free slot or else the victim, the lowest-scored resident not requested in the window (or, if all are, the one requested furthest ahead), unless the window shows the victim needed before the miss; admissions are copied in the step. With no window this is admitting every miss, the online policy that reads least at these budgets; with the whole problem in view it is MIN with one read. *Deployed + window* gives the same window to the deployed policy, so that foresight changes only what is cached. A degraded window keeps each future expert with probability  $r$  and replaces it otherwise, so precision and recall are both  $r$ . The engine plans each step exactly as a replay script does (within 1.7% on the panel’s GPUs).

### Against the deployed cache, a short window pays little.

On the deployed path (Table 5, 4 hosts), a 16-token window runs  $1.00\text{--}1.03\times$  the deployed cache at gpt-oss 11% and  $1.06\text{--}1.12\times$  at 25%, about what MIN with two reads gains; shorter or half-right windows lose up to 6%. Copied in the

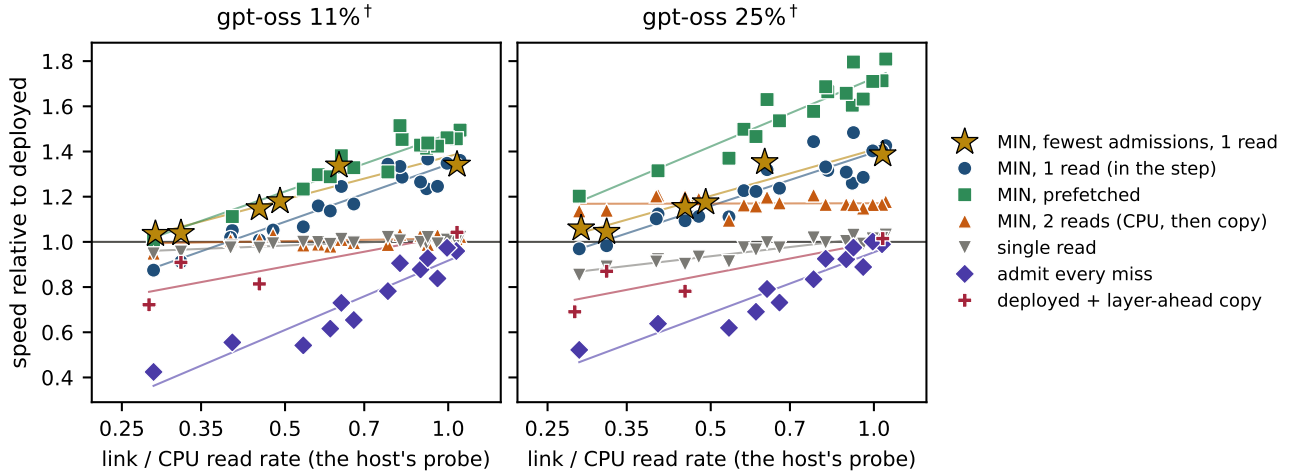


Figure 2. Which way of spending foresight pays depends on the machine: speed relative to the deployed cache on the same host against the host’s link-to-CPU read-rate ratio from its own probe, at the two gpt-oss host-bound budgets, on hosts O3–O6, the panel and the new machines of the follow-ups. Stars: MIN’s fewest-admission set, copied in the step (job 104 and 105, 6 machines). Crosses: the deployed cache with the engine’s layer-ahead copy (job 105). Lines: least-squares fits in the log of the ratio, for the trend only.

Table 5. Foresight on each read path (job 100): speed relative to the deployed cache on the same host. *In step*: admit every miss with a window of  $W$  exact tokens, each admission copied in the step (as on the panel). *Deployed path*: the deployed policy with the same window, its misses split between the CPU and an in-step copy by the host’s fetch table and its admissions copied in the background;  $W=8$  at recall 0.5 in the column marked  $r$ . Hosts sorted by the probe’s link-to-CPU ratio.

| Host               | Link/<br>CPU | Budget      | In step<br>$W=4$ | 16   | Deployed path<br>4 | 16   | $r$  | MIN,<br>2 reads | MIN,<br>1 read | MIN,<br>pref. |
|--------------------|--------------|-------------|------------------|------|--------------------|------|------|-----------------|----------------|---------------|
| Pd again           | 0.41         | gpt-oss 11% | 0.66             | 0.89 | 1.02               | 1.03 | 0.94 | 1.00            | 1.05           | 1.11          |
| Pd again           | 0.41         | gpt-oss 25% | 0.69             | 0.81 | 1.03               | 1.12 | 1.01 | 1.19            | 1.11           | 1.29          |
| 9950X, slower link | 0.54         | gpt-oss 11% | 0.65             | 0.89 | 1.01               | 1.01 | 0.96 | 0.99            | 1.07           | 1.23          |
| 9950X, slower link | 0.54         | gpt-oss 25% | 0.69             | 0.80 | 0.96               | 1.06 | 1.00 | 1.10            | 1.11           | 1.37          |
| 5700X3D            | 0.92         | gpt-oss 11% | 1.09             | 1.32 | 0.99               | 1.00 | 0.95 | 0.99            | 1.37           | 1.44          |
| 5700X3D            | 0.92         | gpt-oss 25% | 1.06             | 1.23 | 1.03               | 1.07 | 1.00 | 1.16            | 1.48           | 1.80          |
| Ph again           | 0.96         | gpt-oss 11% | 0.99             | 1.20 | 1.01               | 1.03 | 0.98 | 1.03            | 1.27           | 1.45          |
| Ph again           | 0.96         | gpt-oss 25% | 0.97             | 1.09 | 1.03               | 1.07 | 1.01 | 1.16            | 1.29           | 1.64          |

step, the same window beats the deployed path where the link reads at least 0.92 of the CPU rate and loses to it at 0.54 or below; on the 10-host panel a 4-token window in the step is slower than the deployed cache on 6 hosts at 11%, a 16-token one on 2. On the deployed path the engine misses 7–33% more than an instant replay (the deployed state matches its replay within 2.4%); background copies that land late are the likely cause.

#### Against admitting every miss, it recovers a share fast.

Figure 3 gives the share of MIN’s gain over admitting every miss that each window recovers. At gpt-oss 11%, exact windows of 1, 4 and 16 tokens recover 0.14–0.17, 0.32–0.47 and 0.78–0.93 of it in time; at 25%, where the cache turns over more slowly, 0.21–0.24 at 4 tokens. The calibrated

model predicts these shares within a median 0.03 (60 host-budget-windows). Time lags reads where the link is slow, because a window still copies 97% of its misses in the step (MIN: 50%); its share of the time gain falls below its share of the read saving by 0.05 where the two paths are equally fast and 0.20 at a ratio of 0.29. Accuracy matters as much as horizon: an 8-token window right half the time recovers 0.20–0.26, about what 2 exact tokens buy, and loses to the deployed cache on 8 of 10 hosts and, on the deployed path, on all 4 (0.94–0.98 $\times$ ; Appendix I).

**The horizon across nine models.** An event-level simulator replays each model’s own sampled text (33,000–152,000 tokens) (Zhang, 2026b). Against admitting every miss, which reads least among LRU, LFU, two decayed-frequency variants, ARC and S3-FIFO in 18 of 27 points (Appendix H), the window  $W_{50}$  that closes half of the read gap to MIN ranges from 0.6 to 34 tokens. Paging theory measures look-ahead in distinct pages (Albers, 1997; Breslauer, 1995), and so does the rule that fits: the experts a layer requests within  $W_{50}$  number about  $0.65 C$  (quartiles 0.61–0.72). Fitted on eight models and tested on the ninth, this one-constant rule predicts the held-out horizon as well as a two-constant power law in tokens (Table 6; the two are not distinguished) and better than  $W_{50} \propto C/k$ . Its value is the unit: a forecaster’s target is the next  $\approx 0.65 C$  distinct experts of each layer.

**Nothing realisable reaches it.** *Online policies* cannot (above). A *learned admission order* reads 6–13% fewer experts in the engine than the deployed policy with single

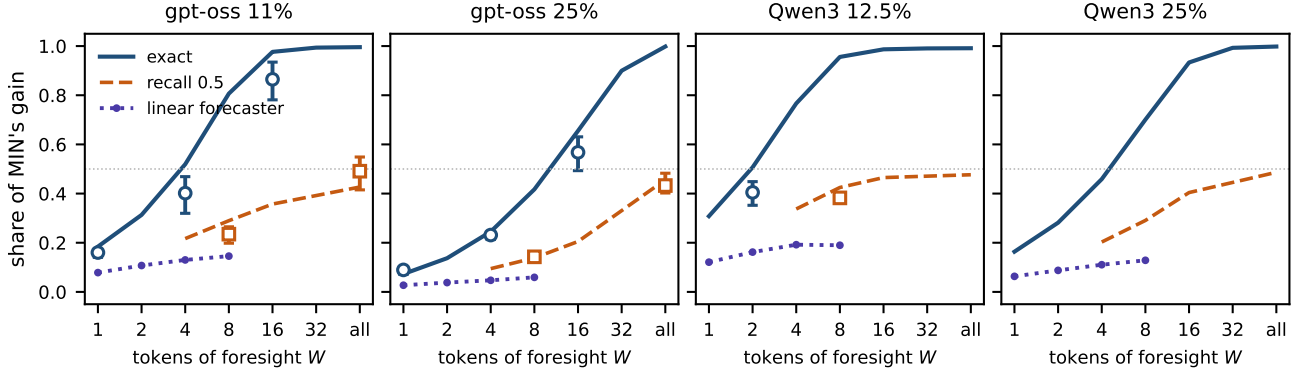


Figure 3. The value of partial foresight at the four host-bound budgets: the share of MIN’s gain over *admit every miss* recovered by exact routing of the next  $W$  tokens (solid), by a window whose experts are each right with probability 0.5 (dashed), and by a linear forecaster fitted on other text (dotted), in host reads on the AIME routing the engine runs; markers: the same share in time per token on the panel hosts (mean, and range over hosts; 10 hosts for gpt-oss, 3 for Qwen3 at 12.5%; no Qwen3 run at 25%). The grey line marks half of the gain.

Table 6. Predicting a held-out model’s horizon  $W_{50}$  (fitted on the other eight models, tested on the ninth, over all 26 budget points): error as a factor,  $\max(\hat{W}/W, W/\hat{W})$ .

| Rule                 | Constants     | Median | 90th pct. | Max  |
|----------------------|---------------|--------|-----------|------|
| $W_{50} \propto C/k$ | 1             | 1.22   | 1.57      | 2.42 |
| $W_{50} = a(C/k)^b$  | 2             | 1.19   | 1.51      | 1.67 |
| $D(W_{50}) = cC$     | 1, and $D(W)$ | 1.16   | 1.42      | 1.61 |

reads but runs  $0.90\text{--}1.05\times$  the deployed cache. *Speculative batches* save an online cache 8–12% of its reads at 90% draft acceptance and cost 18–20% at 80% (Chen et al., 2025b; Li et al., 2026; Saxena et al., 2026). *Layer-ahead predictors* (Hwang et al., 2024; Zhu et al., 2025) reach 84% recall in our engine but see less than one token ahead, so they can only hide reads under the GPU’s compute. The engine’s layer-ahead copy spends them that way, copying one predicted expert per layer ahead of use (84% of the copies are used): at gpt-oss 11% it gains  $1.04\times$  on the machine whose link is as fast as its CPU and loses on the other 3, down to  $0.72\times$  (job 105, Table 4). The gain, and the losses on the 2 machines whose link reads at under 0.4 of their CPU rate, were predicted from that ratio before the runs; the third loss, on a machine between the registered classes, was not. *Forecasters* trained on routing histories, linear or recurrent, close at most 15% of the read gap on gpt-oss: their precision falls with the horizon (0.48 one token ahead, 0.26 four ahead), and over a window they do no better than random errors of the same precision (Fig. 3, dotted; Appendix I).

## 6 RELATED WORK

**Offloading systems.** Demand-paging systems cache experts per layer and copy misses over PCIe (Eliseev & Mazur, 2023; Xue et al., 2024; Song et al., 2024a; Tang et al., 2024; Kong et al., 2024; Zhong et al., 2024; Yi et al., 2023; He et al., 2024; Shams, 2026), or use predicted expert activations to decide what to keep (Du et al., 2024); an upstream llama.cpp pull request adds such a cache, admitting every miss into an LRU (Gupta, 2026). CPU-execution systems run non-resident experts on the host (Kamahori et al., 2024; Chen et al., 2025a; Zhong et al., 2025; Zhang et al., 2025; Wang et al., 2026a; ATSIInfer), as PowerInfer runs a dense model’s cold neurons (Song et al., 2024b); a set-associative GPU cache with CPU-executed misses (Huang et al., 2025), a llama.cpp fork (leloch, 2026a;b) and two llama.cpp pull requests (miltos22, 2026; csantiago78, 2026) are the closest designs to ours. FreeToken (Yang et al., 2026) and DALI (Zhu et al., 2026) split each layer’s misses between CPU and PCIe. Predictors guess upcoming experts from the previous layer’s state (Hwang et al., 2024; Zhu et al., 2025), several tokens ahead with a learned forecaster (Wang et al., 2026b; Yadav & Asgari, 2026), or from draft tokens (Chen et al., 2025b; Li et al., 2026; Saxena et al., 2026); Read-ME decouples the router so routing is known in advance (Cai et al., 2024). Batched-throughput systems (Sheng et al., 2023; Cao et al., 2024; Yuan et al., 2025) and flash or SSD tiers (Alizadeh et al., 2024; Liang et al., 2026a; Lin et al., 2026) are outside our scope.

**Bounds, oracles and caching theory.** WiSP (Zhang et al., 2026b) bounds decode by PCIe turnover, Budgeting Bytes (Zhang, 2026a) by bytes over bandwidth for a given fetch set, and MoE-CAP (Jiang et al., 2025) measures bandwidth

utilisation against a single peak; the roofline (Williams et al., 2009) is the single-memory ancestor of our bound. Zhang (2026b) proposes a trace-level evaluation contract and reports a 44–46% gap between causal policies and MIN with bypass; Liang et al. (2026b) measure, across 20 models, the hit rate of an oracle cache that sees  $m$  tokens ahead against the cache ratio  $C/k$ . We add the time domain: an oracle inside a real engine, the cost of how foresight is spent, and a horizon rule in distinct experts. SAEM (Zhang et al., 2026a) gives its own stage-level planner perfect knowledge of the coming stage in its engine and finds that worth at most about 14% on an A100; SpecMD (Hoang et al., 2026) benchmarks expert-cache and speculative prefetch policies across hardware; Angelopoulos et al. (2025) give competitive bounds for layered expert paging. With bypass, MIN maximises hits exactly (Carlisle & Lloyd, 1995; Jain & Lin, 2016); when bypass and admission cost differ, the time-optimal placement is a different problem (Zhang et al., 2020), which is why we use MIN’s reads as an input to a bound rather than as a time-optimal policy. Choosing among hit-optimal schedules by what admissions cost has precedents: Demand-MIN for prefetching (Jain & Lin, 2018), offline-optimal caching as a min-cost flow over reuse intervals (Berger et al., 2018), and admission rationing in flash caches (Eisenman et al., 2019). Our fewest-admission schedule is that interval formulation with admissions as a second objective, measured in an engine. Lookahead in paging is valuable when measured in distinct pages (Albers, 1997; Breslauer, 1995), and learning-augmented paging is most robust with predictions of whether a page will be evicted before its next use (Antoniadis et al., 2023); learned replacement imitates Belady in CPU and content-delivery caches (Jain & Lin, 2016; Song et al., 2020) and in an SSD-backed expert cache (Kim et al., 2026).

## 7 LIMITATIONS

- **Hardware and sample.** Rented consumer cards in desktop hosts; clocks could not be locked and power limits differed. The panel has 10 hosts, one launch each (4 machines relaunched later); the full  $2 \times 2$  ran at every budget on two hosts. Few hosts have a link slower than half their CPU rate, and the greedy copy loses on 2 machines only, so the crossover in Fig. 2 rests on few points; the line at half was drawn after the panel ran. The fewest-admission schedule ran on 6 machines and the layer-ahead copy on 4 machines, both at two budgets; the two machines with the slowest links differ in other ways too (one runs its card at  $\times 8$ , the other at  $\times 16$  behind a much faster CPU path). Within-host intervals resample problems, which carry little of the variance; launch-to-launch differences are of the size of the smallest gains we report, which is why we count machines and relaunched the ones that

decide a claim.

- **Workload.** Batch-1 decode of two models on AIME prompts, also used while developing the system, teacher-forced on each model’s own greedy text. Prompts and batches above one bypass the cache.
- **The bound.** Exact routing and per-layer slots (pooling slots across layers lowers MIN’s reads by 4.5–18.6%). It drops latencies and lets the read paths share the best probed rate, so a faster reader than our probe could exceed it: on one launch the engine’s reads, at the profiled  $G$ , ran faster than the probe’s highest rate. The GPU term of Eq. (2) is our engine’s kernel time, not a property of the machine alone. Its read time is nearly reachable in a microbenchmark that runs no compute; whether the GPU’s work overlaps the reads as the bound assumes we have not measured.
- **The oracles and the model.** Oracles read a recorded trace. In the engine the fewest-admission schedule misses 2.5–5.3% more than the greedy one, more than its replay predicts; its measured gains include that cost. Equation (3) has no term for background copies that overlap the GPU’s work: at gpt-oss 25%, where they are a larger share of the reads, it over-predicts by up to 18% and failed its registered test; at 11% it misses 2 of 30 launches by more than 6% (–10 and 14%). It was tested on gpt-oss only. The two-path model is optimistic for copies in the step on slow links and for background copies, and it fails on a 12-channel server host (Appendix C). Degraded windows draw errors independently; a real forecaster’s errors are correlated with what the cache holds.
- **Outputs.** Parity is measured by teacher-forced KL divergence against an all-VRAM reference (mean 0.0019 and 0.0005 nats), not by task accuracy.

## 8 CONCLUSION

With experts in host memory, batch-1 MoE decode is the GPU’s compute plus host bytes, in series for a system that reads on demand, apart from the copies it makes in the background. Our cache executes its reads at the machine’s best rate, so what separates it from the bound is the serialisation of the GPU’s work with the reads and the reads it makes beyond MIN’s. Knowing the future reduces both when it is spent as MIN spends it, caching MIN’s set and reading each expert once, and the choice among MIN’s hit-optimal schedules matters: the fewest-admission one gains on every machine it ran on, including the slow-link machines where the greedy one loses. How much it gains is a property of the machine that its read rates largely predict. And foresight has a measurable price: about  $0.65 C$  distinct experts of each layer’s coming routing to close half of the read gap between admitting every miss and MIN, at accuracies no

forecaster we know of or trained reaches, while one layer of foresight, which predictors already provide, can only hide reads under the GPU’s own compute and, spent by the engine’s layer-ahead copy, pays only where the link is as fast as the CPU.

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## A SCIENTIFIC BENCHMARKING CHECKLIST

How this paper follows the twelve rules of [Hoeffler & Belli \(2015\)](#).

| Rule                                                                         | Where and how                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Report the base case of a speed-up and its absolute performance.          | Every ratio in Table 2 is next to both absolute speeds; llama.cpp is configured with per-layer expert offload at equal expert memory, the setting its users run.                                                                                                                                       |
| 2. Explain subsets of benchmarks or resources.                               | Two models chosen for size and precision (MXFP4, BF16), a third that is FreeToken’s own headline model; budgets span what fits a 32 GB card; the 43.75% budget is omitted where it does not fit. All 30 AIME-25 problems are used.                                                                     |
| 3. Arithmetic mean for costs, harmonic mean for rates.                       | The accounting, the panel and the value of foresight average time per token over problems (a cost) and quote speeds as the harmonic mean this implies; in Table 2 every problem decodes the same number of tokens, and ratios of total time agree with the reported ratios of mean rates within 0.003. |
| 4. Avoid summarising ratios.                                                 | Ratios of means are never averaged across budgets or models; ranges list the per-budget values, and the medians reported across models (Table 15) or across published rows (Appendix K) are medians of per-cell quantities, labelled as such.                                                          |
| 5. Report whether measurements are deterministic; give confidence intervals. | Decode is greedy and deterministic per engine; timing is not. Every comparison has a 95% interval over problems; the panel (job 099) adds intervals over hosts, the unit that varies most.                                                                                                             |
| 6. Do not assume normality.                                                  | Intervals are bootstrap percentile intervals (10,000 resamples, paired by problem); no distributional assumption.                                                                                                                                                                                      |
| 7. Compare nondeterministic data soundly.                                    | Paired bootstrap of the ratio of means within a host; across the panel’s hosts, a two-stage bootstrap (hosts, then problems) ( <a href="#">Kalibera &amp; Jones, 2013</a> ); the variant a system is scored with is picked on a separate launch (except the Qwen3 25% row of Table 2).                 |
| 8. Check that central tendency is the right summary.                         | Mean speed over 256 tokens is FreeToken’s own metric; per-token p50 and p99 latencies are recorded for every problem in the released rows.                                                                                                                                                             |
| 9. Document all factors and the setup.                                       | Machines, probes, power limits, memory clocks (Section 2), software commits (llama.cpp 4da6337, FreeToken 0d652e7), flags and job scripts are released; Appendix L.                                                                                                                                    |
| 10. Report measurement, synchronisation and summarisation.                   | One client times every system of Table 2 from streamed token arrivals; the oracle comparisons of Section 4 are timed in process by llama-ec-bench over teacher-forced text, every policy by the same timer (Section 2).                                                                                |
| 11. Show upper performance bounds.                                           | The bound (Section 3) is a column of Table 2; Table 18 tightens it four ways; Section 3 applies it to the published literature and to our cache with the same procedure.                                                                                                                               |
| 12. Plot enough to interpret; connect points only for trends.                | Figure 1 plots every host’s ratio with its interval; Fig. 3 connects measured windows for the trend only and marks engine measurements separately; Figs. 6 and 8 are bar charts and Figs. 4, 5 and 9 scatter plots.                                                                                    |

## B PREDICTION LOG

Each job’s predictions are in the header of its script on the public `gpu` branch, committed and pushed before the machine started; git history orders every prediction before its measurement. Outcome notes with the statistics are in `prereg/`. The scorecard, generated from `prereg/scorecard.clauses.json`, splits every prediction of jobs 073–098 into its separable clauses and scores each under one rule (jobs before 073, which built and debugged the engine, also registered predictions in their headers; their outcome notes are in `prereg/`, and they are not scored here; job 072’s, for one, failed or did not apply on two of its three): *held* only when the point estimate is on the predicted side and the 95% paired-bootstrap interval excludes the threshold (for a band, lies inside it); *held (point)* when only the point estimate is, or no interval exists

(except in the hand scoring of jobs 073–098, which holds deterministic counts such as reads outright, since they have no sampling error); *failed* when the point estimate is on the wrong side; *untested* when the clause could not be evaluated; *void* when the job was voided. Of 565 clauses, 218 held, 193 held on the point estimate, 142 failed, 8 were untested and 4 void: 7 of 19 scored clauses held in jobs 073–075, when the system was being built, 19 of 23 in jobs 076–081, 11 of them sign-only predictions made after a same-host pilot, 24 of 38 in jobs 082–087, and 168 of 473 in jobs 088 onward, where 42 of 82 sign clauses held (75 on the point estimate) and 90 of 301 bands failed. Most bands that failed are the sizes of effects whose direction held: our slow-link lead landed at  $1.51\text{--}1.58\times$  against a predicted  $1.10\text{--}1.50$ ; the rerun of Table 2 on a second host (job 089) gave ratios  $0.06\text{--}0.11$  below host B’s at five of six cells against a predicted  $\pm 0.06$ , host-bound speeds  $14\text{--}17\%$  lower against a predicted  $\pm 12\%$ , and LRU  $6\text{--}8\%$  slower than decayed frequency at the two GPU-heavy cells against a predicted  $10\text{--}30\%$ ; on the 4090 and 3090 hosts (jobs 091, 092; Appendix D) our lead over FreeToken and its fraction of the limit overshot their bands, FreeToken’s carried-over backend fell far below its band, and two launch-order checks exceeded  $3\%$  ( $7\%$  and, at gpt-oss  $11\%$  on the 3090 host,  $11\%$ ); in the first oracle job (093; Section 4) the oracle’s gain fell outside  $20\text{--}80\%$  at the two lowest budgets (it lost), its speed fell just below the predicted  $45\text{--}65\%$  of the limit at all four host-bound cells ( $43.7\text{--}44.8\%$ ), its hit rate and admissions missed the optimum’s by more than the predicted 3 points and  $30\%$  at three and two of the four, no-overlap cost  $0\%$  against a predicted  $5\text{--}25\%$ , and all-CPU cost  $15\%$  on Qwen3 against a predicted  $0\text{--}10\%$ ; in the bytes-oracle job (094) the bands missed as Appendix G describes; in the single-read job (095) 44 of 56 clauses held, and the 12 that failed are of size: the model on the fetch oracle’s counters was predicted within  $10\%$  and missed by  $13\text{--}17\%$  at four cells, the fetch oracle’s reads were predicted within  $10\%$  of the simulation’s and came  $17\%$  below at the two GPU-bound cells (the registered simulator left a resident carried over from the previous sequence with no next use, and so evicted it first; corrected, the simulation is within  $4\%$  of the engine’s reads at every cell, Appendix G), the prefetching oracle was predicted at  $55\text{--}85\%$  of the limit at the host-bound cells and reached  $44$  and  $51\%$  at the two  $25\%$  cells, the model’s over-prediction of it was predicted at  $15\text{--}50\%$  and came to  $7$  and  $13\%$  at the two lowest budgets, and the paced variant, predicted within  $10\%$  of the unpaced one, was  $12\text{--}13\%$  faster at the two  $25\%$  cells; in the confirmatory factorial (096) the clauses held at most cells of both hosts and the failures are listed in Appendix G, the largest being the difference between the hosts; and in the FreeToken tuning job (098) its best setting stayed within the predicted  $10\%$  of its default and our ratios within the predicted bands, while the fixed fetch caps, the thread count and the CPU-only comparison missed as `prereg/freetoken_outcome_098.md` describes. Among the 23 threshold failures are ours at  $1.56\text{--}1.78\times$  llama.cpp on the slow hosts against a predicted  $1.8$ , the Mixtral tie against a predicted  $1.3\times$ , the hit-optimal oracle’s gain at the two GPU-bound cells ( $48$  and  $56\%$ ) against a predicted ceiling of  $15\%$ , the bytes-optimal oracle short of beating the hit-optimal one by  $15\%$  at the two lowest budgets, and its 16-token window short of keeping  $70\%$  of its gain at three cells, fetch’s gain differing between hosts O4 and O5 by more than  $0.15$  at five cells, Belady’s single-read prefetch above  $1.05\times$  the online policy at Qwen3  $12.5\%$  on O4, and FreeToken’s CPU backend running close to stock llama.cpp’s where we predicted at least  $2\times$  and  $1.2\times$ ; the 3 equality failures are two of the 4090 host’s FETCH tables, which copy one more expert than predicted, and FreeToken’s best backend at gpt-oss  $25\%$  in job 098 (offload, not hybrid); the 7 sign failures are LRU behind FreeToken at gpt-oss  $11\%$ , where we predicted it would still lead, and the hit-optimal oracle slower than the online policy at gpt-oss  $11\%$  and Qwen3  $12.5\%$ . Against the running log kept during the study, two clauses are downgraded (a llama.cpp clause of job 074b that the log counted as held was never run; job 085b’s own restatement of its fourth prediction, “at least 2 of 3”, failed at 1 of 3, though the pooled 4 of 6 held), job 073’s “identical outputs” is scored on fully identical outputs ( $0\text{--}3\%$ ) rather than on agreement of the first 160 characters ( $27\text{--}47\%$ ), and ten clauses the log counted as held, among them job 087’s first, hold on the point estimate only under the interval rule; `prereg/scorecard_outcome.md` lists every change. Job 085’s Qwen3 half ran as 085b on a second machine with a slow link after a hung conversion. Job 084 ran on a throttled card and was rerun, with its predictions unchanged, as 084b/084c. The batched-verification variants of Section 5 (rejected drafts free, routed like the true token, or routed at random) are in `prereg/batchchk_outcome.md`.

The clauses were scored by hand from the outcome notes. A script (`scripts/scorecard_auto.py`) re-scores every clause whose measured value is a number and whose threshold parses, 469 of 565: it agrees with the hand score on 432; of the 37 differences, 36 are the deterministic counts (reads, divergences, table entries) that the hand scoring holds outright and the script scores held (point), and 1 is a boundary value the hand scoring read unrounded ( $1.249$  against a band starting at  $1.25$ ). Of the clauses the script scores held on the point estimate, 213 have no interval and 7 have one that crosses the threshold. Job 099’s clauses are scored by machine from the start (`scripts/panel_099.py`): of 273, 190 held, 42 held on the point estimate, 39 failed and 2 were untested (`prereg/scorecard_099.json`); the two untested clauses are one host’s replay check, whose step failed when a Python package did not install. The failures concentrate on the two hosts whose link reads at less than half their CPU’s rate, where the bands, set from four earlier hosts none of which was link-starved, did not encode host dependence; and on the planning time of the noisy whole-future window (`prereg/panel_outcome_099.md`). Job

100 (scripts/job100.py): of 216, 41 held, 122 held on the point estimate, 27 failed and 26 were untested. Most failures are one prediction, that the deployed-path windows miss within 10% as often as a replay that admits at once (their copies land late); the rest are the probe-only model on the half-width-link host and four band or ordering edges. The untested clauses are the planning time of the deployed-path windows, which the engine does not time, and the paired prediction for a host that never reached a timed step (prereg/window\_outcome\_100.md). Job 101 (scripts/job101.py; the lowest-ratio panel machine relaunched, and a second Ryzen 7 9800X3D behind a slower link): of 25, 6 held, 19 held on the point estimate, 0 failed and 0 were untested (prereg/crossover\_outcome\_101.md). Job 102 (scripts/job102.py; the read-schedule microbenchmark on three hosts): of 33, 0 held, 28 held on the point estimate (none has an interval), 5 failed and 0 were untested. Four failures are one prediction, that waiting per layer would cost at least 0.03 of the host term against the analytic schedule; on two of the three hosts it costs nothing measurable. The fifth is the ordering of the two budgets on the Intel host (prereg/readsched\_outcome\_102.md). Rentals that produced no results were destroyed and are in the rental ledger (gpu/vast\_ledger.json): the first two for 099a, replaced before any measurement; 100d, which stalled downloading the model; 100e, which never reached its job; and 103c, which never started its job. Job 103 (scripts/job103.py; MIN’s fewest-admission schedule against the greedy one): the hosts could not install the plan’s solver beside Debian’s numpy, so the engine ran the greedy schedule in the plan configurations, as it logs; of 45 clauses 4 held on the point estimate (the relaunch of Pf) and 41 are untested. Job 104 repeated it with the plan in its own environment on three of the same machines (Pf, a second Core Ultra 9 285K whose link reads at 0.49 of its CPU rate, and Pg, whose second launch job 103’s 5950X turned out to be): of 47, 1 held, 22 held on the point estimate and 24 failed. Most speed predictions held (the fewest-admission set gains on Pf, and gains at least 0.03 over the greedy set at gpt-oss 11% on every machine; with two reads it is at most 0.01 behind the greedy set at 5 of 6 cells, and 0.047 behind on Pg at gpt-oss 25%); the other failures are the counters at gpt-oss 25% (copies 0.73 of greedy’s against at most 0.70, misses 5% more against at most 3%), the engine’s misses against the host’s replay, the two-read reads, and the interaction, which we predicted would shrink with the plan’s set and which grew (prereg/minadm\_outcome\_104.md). Job 105 (scripts/job105.py; Eq. (3) with  $G$  profiled on each host before its timed runs, the engine’s layer-ahead copy, and the fewest-admission set, on an EPYC 7402P, a Threadripper 9960X, Pf and O4): of 31, 7 held, 20 held on the point estimate and 4 failed. Every failure is the law at gpt-oss 25%, which over-predicts the deployed cache by 6.9–9.3% (against at most 6%) on 3 of the four hosts (3.8% on Pf) and so takes the pooled median error (4.2%) past the 4% registered; at 11% it is within  $-1.5$  to  $4.7\%$ . Every speed prediction held: the layer-ahead copy gains at least 3% on the host with a ratio above 0.8 and loses on both below 0.4, and the fewest-admission set beats the deployed cache on every host, beats the greedy set by at least 0.03 on the three with a ratio under 0.7, and copies at most 0.65 of the greedy set’s experts (prereg/sumlaw\_outcome\_105.md). Two clauses predicted quantities already measured, the profiled  $G$  (job 069c) and the plan’s copies, which are a property of the trace (job 104); they test that the engine reproduces them. Three more launched hosts produced no results: 105c never started its job, and the job’s bandwidth gate stopped 105d and 105g, whose cards were throttled, before any timed run.

The clause-by-clause tables (565 rows for jobs 073–098, 273 for job 099, 216 for job 100, 25 for job 101, 33 for job 102, 45 and 47 for jobs 103 and 104, 31 for job 105) are in the supplement (paper/supplement.pdf), generated by the same scripts.

## C THE CALIBRATED TIME MODEL

Our cache’s one per-machine decision, how many of a layer’s misses to fetch rather than run on the CPU, comes from a calibrated model of time per token. With  $X_c$  the bytes the CPU reads per token and  $X_p$  the bytes the link reads,

$$T = G + \max\left(\frac{X_c}{B_c}, \frac{X_p}{B_p}, \frac{X_c + X_p}{B_{cp}}\right), \quad (4)$$

with one constant  $G$  per engine and model, fitted once on one host and frozen. It is a calibrated model, not a law:  $G$  is fitted, and it exceeds the whole all-in-VRAM token (5.17 against 3.92 ms) because it absorbs per-token host work and the CPU helpers’ shortfall against the probe (below). As a predictor of time it ties a naive one-path model with the same  $G$  (time =  $G$  + all host bytes over the best probed rate):

| Held-out set                     | Eq. (4) |           | one path |           |
|----------------------------------|---------|-----------|----------|-----------|
|                                  | median  | 90th pct. | median   | 90th pct. |
| blind test, 33 rows on 7 hosts   | 3.4%    | 12%       | 3.6%     | 31%       |
| frozen, 30 rows on 4 later hosts | 5.6%    | 16%       | 6.1%     | 11%       |

Table 8. The calibrated model predicted blind: each machine wrote its prediction from its own bandwidth probe, with constants frozen in the public repository, before any model run. Error of predicted over measured speed, gpt-oss-120b (cache rows) and llama.cpp --n-cpu-moe (last column, %).

| Host CPU                   | Job  | $n$ | Median<br> err  % | Max<br> err  % | llama.cpp<br>err % |
|----------------------------|------|-----|-------------------|----------------|--------------------|
| Core Ultra 7 270K (8P+16E) | 069c | 5   | 1.5               | 11.9           | +119               |
| EPYC 7352 (Zen 2)          | 069c | 5   | 3.2               | 11.7           | +10                |
| EPYC 9655 (12-ch. DDR5)    | 069c | 5   | 18.7              | 31.9           | +16                |
| Ryzen 9 7900               | 072  | 5   | 4.9               | 8.2            | +4                 |
| Ryzen 9 9950X3D (A)        | 073  | 8   | 1.6               | 4.5            | —                  |
| Ryzen 9 7950X              | 074  | 1   | 4.0               | 4.0            | —                  |
| Ryzen 9 9950X              | 076  | 4   | 2.5               | 5.0            | —                  |
| Ryzen 9 9950X3D (B)        | 082  | 0   | —                 | —              | +4, +4             |

It over-predicts Qwen3’s speed on the later hosts by +5 to +21%, a budget-dependent bias whose  $G$  was borrowed from llama.cpp. Its value is the per-machine fetch table it picks (Eq. (5) below), which gains 3.3–8.0% over a fixed table on host B and 9–34% on hosts with half the link (Appendix J).

**The blind test in full.** On every machine of the blind test the job wrote the model’s prediction from its own probe before any model ran (Table 8; all of its cache rows are gpt-oss-120b on RTX 5090 hosts). Over 29 configurations on 7 machines (33 measurements; one machine ran twice) the median error is 3.7% per configuration; without the 12-channel EPYC 9655 it is 3.3% over 24 configurations (90th percentile 7.5%). llama.cpp --n-cpu-moe is predicted within 4% on the Ryzen hosts, including two Qwen3 configurations on host B (+3.6%, +4.0%; for Qwen3,  $G=4.3$  ms, the mean residual of three llama.cpp runs on another host), and misses by +10–+119% on the hybrid-core and server hosts. With its constants frozen, the model scores the cache on the 4 RTX 5090 hosts rented later (30 measurements at the six cells of Table 2, not written before the runs) at a median 5.6%, 90th percentile 16%, worst 21% (Qwen3-30B-A3B 43.75% on oracle host O1). Refit under leave-one-host-out validation, the token-level max has a median error of 3.1% (90th percentile 9.4%), the additive form  $G + X_c/B_c + X_p/B_p$  4.5% (11.3%) and a per-layer max summed over layers 3.2% (10.2%); the max beats the sum on the three rows that copy misses on demand behind a 53 GB/s link, where both paths carry most of the bytes, and on the EPYC 9655 rows, where both fail, the sum is closer on the two rows that copy behind a slow link, and elsewhere the two are within a few points (Appendix C). Each failure has a measured cause (Appendix C): on-demand copies behind a slow link share DRAM less evenly than the probe’s combined rate says; llama.cpp splits every matrix evenly across threads and runs at 46% of the model on a hybrid-core host; and on a 12-channel EPYC host memory stops being the limit and a per-layer latency floor takes over.  $G$  (5.17 ms refit) exceeds the whole all-in-VRAM token (3.92 ms) because it absorbs a shortfall the probe does not see: the helpers read at  $1.09\text{--}1.31\times$  the probe’s time per expert, and a fitted rate factor is collinear with  $G$  at this sample size.

**What the model says, and the split it picks.** Equation (4) is a calibrated model of execution that reads host memory in sequence with its GPU work, as our cache and llama.cpp do. For such an engine every mechanism acts only through the bytes read from host memory per token and the number of read paths in use: a second path adds bandwidth on every host of the blind test but the 12-channel EPYC ( $B_{cp}$  against  $\max(B_c, B_p)$ ): 8–17% on its Ryzen hosts, 1% on the Core Ultra, 33% on the Zen 2 EPYC; the RTX 3090 host’s Core i9-11900KF, where the two paths together read less than the CPU alone, is a second exception), and that is the whole gain available to fetching or splitting misses between the CPU and PCIe. Prefetching never reduces the bytes; what it can buy is overlap of the reads with the step, which the model does not price: charging every read in sequence, it over-predicts the time per token of MIN prefetched by +1 to +38% (Section 4) and of a

Belady prefetch by up to 53% (Appendix G). The model’s job in the system is the split. Its per-layer form,

$$T(n, f) = \max\left(\frac{fS}{B_p} + g, [n > f](L_c + \frac{(n-f)S}{B_c}), [0 < f < n] \frac{nS}{B_{cp}}\right), \quad (5)$$

with  $g$  the GPU’s expert work per layer ( $48 \mu s$  for Qwen3,  $37 \mu s$  for gpt-oss, from profiles) and  $L_c=20 \mu s$  the hand-off, picks for each count  $n$  of missed experts the number  $f$  to copy with the smallest  $T$ . On host B the model’s table beats the fixed table we used before at every budget of both models, by 3.3–8.0%; on hosts whose link we measured at 27–30 GB/s instead of 53 it fetches less and gains 21–34% on gpt-oss and 9–18% on Qwen3, the per-machine split mattering most where the link is slow (Appendix J). In the running example on host B the table is 0, 0, 1, 1, 2 ( $f(0..4)$ ): one missed expert runs on the CPU, hidden behind the GPU’s other experts; from two misses the model copies one. The lead over FreeToken at that cell,  $1.27\times$ , is mostly the admission policy (77% of it by the LRU ablation of Appendix J), the rest the CPU path that FreeToken’s bandwidth model declines to use there.

**Is the max the right form?** In 19 of the 29 configurations nothing is copied, so they test one roof. We therefore refit the model and its alternatives on the same 33 measurements under leave-one-host-out validation (each of the 7 hosts predicted with constants fitted on the other six): the token-level max of Eq. (4) with  $G$  refit has a median error of 3.1% (90th percentile 9.4%); the additive form  $G + X_c/B_c + X_p/B_p$ , 4.5% (11.3%); a per-layer max summed over layers from the engine’s per-layer miss histograms, 3.2% (10.2%). The additive form ties in sample but misses the three rows that copy misses on demand behind a 53 GB/s link, where both paths carry most of the bytes, by -11%, while the max is within  $\pm 0.2\%$  there; on the slow-link copy rows the additive form is the closer of the two. The per-layer form neither beats the max nor explains its failures: the Jensen gap between a sum of per-layer maxima and the maximum of sums is a fifth of the Core Ultra’s residual, none of the EPYC 7352’s, and has the wrong sign on the EPYC 9655 and on the prefetch rows. We keep Eq. (4), with its evidence stated: the max beats the sum on the three fast-link copy rows and on the EPYC 9655 rows, where both fail, the sum is closer on the two slow-link copy rows, and elsewhere the two are within a few points, so the form is settled only where both paths carry most of the bytes. FreeToken’s own runs are not predicted, because its counters are not exposed; the model covers our engine and llama.cpp.

**Where it fails, and why.** Each failure now has a measured cause. (i) Copying misses on demand (FETCH) at 25% is over-predicted by +11.7% on the Core Ultra 7 and +11.4% on the EPYC 7352 (links of 36 and 26 GB/s); +8.8% and +8.5% with  $G$  refit. The engine’s per-layer timers show why: a CPU miss served beside a copy takes 20–62% longer than one served alone, so the two paths share DRAM less evenly than the probe’s combined rate says; a variant of the model that uses the probe’s per-path rates under contention and a per-layer hand-off constant predicts both rows within +3.1% and +4.1%, at a worse median (4.2%). (ii) llama.cpp on a Core Ultra 7 with 8 performance and 16 efficiency cores runs at 46% of the model’s prediction at its default thread count, because it splits every matrix evenly across its threads; our helpers claim work dynamically and stay on the model on the same machine. We did not test llama.cpp restricted to the performance cores. (iii) On a 12-channel EPYC 9655 (559 GB/s) host memory stops being the limit (+16 to +29% with  $G$  refit under leave-one-host-out validation; Table 8 gives the blind errors): the per-layer timers show a floor of 55–70  $\mu s$  per layer with CPU misses that no desktop host has (their intercepts are at or below zero), the latency of the hand-off at a bandwidth where a 13 MB expert takes 24  $\mu s$ . (iv) Prefetch configurations are under-predicted by -8.7 to -3.4%: their copies overlap attention, which the model charges in full.

## D THE GRID: OTHER CARDS, AND THE SCOPE OF THE GAIN

Table 9 lists every cell of Table 2’s protocol on the four machines we measured, each against its own limit, and Fig. 4 plots the fractions. The protocol was repeated on an RTX 4090 next to a Core i5-12400 (the host reads 37 GB/s, the link 25) and an RTX 3090 next to a Core i9-11900KF (45 GB/s; CPU and link together read *less* than the CPU alone, so the model’s table copies nothing), at the cells that fit 24 GB. On the 4090 ours runs  $1.03\text{--}1.37\times$  FreeToken and  $1.8\text{--}2.6\times$  llama.cpp at half host B’s absolute speeds, the host being the reason; on the 3090 it runs  $1.6\text{--}2.2\times$  llama.cpp (FreeToken’s carried-over backend ran below llama.cpp there and is not counted). The cards’ own bandwidth never enters: every one of the seven cells on the 24 GB cards is host-bound at the limit, and ours stands at 30–46% and 27–42% of it. Mixtral-8x7B (8 experts per layer, top-2; 26 GB of experts on the 3090) marks the scope: with 2 of 8 experts resident the cache hits 31% of reads where whole-layer pinning hits 25%, so there is little locality to earn, and ours ties llama.cpp (1.015 [1.008, 1.024]); on the 128-expert models the same cache hits 53–79% at 11–25% resident. Across eight hosts the gpt-oss lead over FreeToken grows with the host’s CPU-to-link ratio, from  $1.09\text{--}1.22\times$  where the CPU reads  $1.05\text{--}1.14\times$  its link’s rate to  $1.51\text{--}1.58\times$

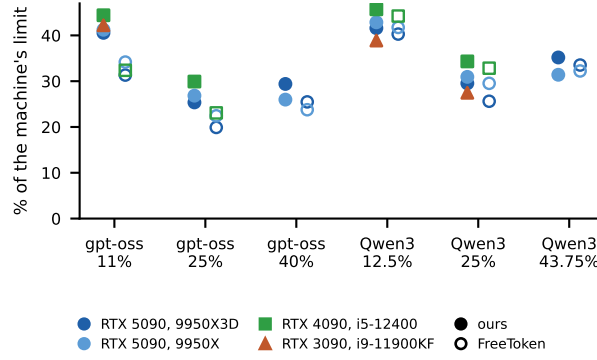


Figure 4. Ours (filled) and FreeToken (hollow) as a percentage of each machine’s own bound (exact optimum, the machine’s highest probed host rate, the card’s datasheet rate) at every cell of Table 9: two RTX 5090 hosts, an RTX 4090 and an RTX 3090 (FreeToken’s 3090 cells, its carried-over backend running below llama.cpp, are omitted). The limit ranges from 41 to 519 tok/s across these cells; ours stays at 25–46% of it.

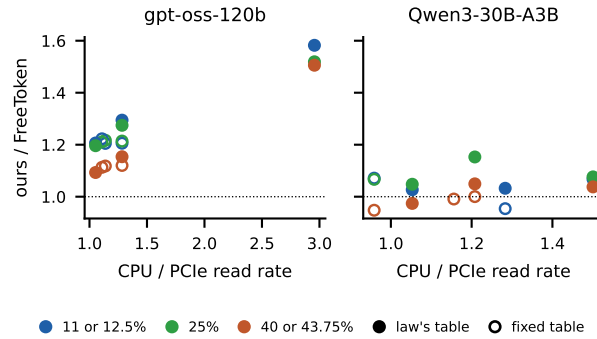


Figure 5. Ours over FreeToken’s better backend (carried over from host B on host S) in every same-machine comparison of the current engine on the two models of Table 2 (eight hosts), against the machine’s CPU-to-PCIe read ratio (FreeToken’s own probe). Filled: FETCH table from the model on that machine; hollow: the fixed table. The rightmost gpt-oss points are a Core i9-14900K with FreeToken’s executor on its performance cores; the rightmost Qwen3 points are a Ryzen 9 7900.

behind a link of half the bandwidth, in the direction the model’s split implies (the model does not predict FreeToken’s runs), and the Qwen3 lead stays at  $0.95\text{--}1.15\times$  (Fig. 5, Appendix J).

The 24 GB cards run the cells that fit (the VRAM arithmetic, written on the machine from the GGUF header, skipped Qwen3 at 43.75%: 28.0 GB of slots, dense weights and KV against a 23 GB limit). The predictions for jobs 091 and 092 and their outcomes are in `prereg/grid_outcome_091.md` and `092.md`: on the 4090 our lead over FreeToken at gpt-oss 11% (37%) exceeded the predicted 10–35% band, ours ran  $1.78\times$  llama.cpp at Qwen3 12.5% against a predicted 1.8, the model’s tables copied one more expert than host B’s at one entry each (the prediction reasoned from the link alone; the CPU is slower in the same proportion), and ours’ fraction of the limit at Qwen3 12.5% (45.7%) exceeded the predicted 25–45%; on the 3090 FreeToken’s carried-over backend ran below llama.cpp, ours ran  $1.56\text{--}1.64\times$  llama.cpp at two cells against a predicted 1.8, Mixtral tied at  $C=2$  against a predicted  $1.3\times$  and its  $C=4$  cell was lost to the job’s deadline, and one launch-order check exceeded 3% (7%).

## E THE ORDER-FREE ACCOUNTING OF THE GAP

Two choices separate the deployed cache from MIN with one read: *what to cache* (the deployed policy’s admissions or MIN’s) and *how to load* (by the CPU, two reads, or in the step, one). The engine ran all four combinations on O4, O5, the panel and the three new machines of jobs 100–101. In the running example, changing only what to cache closes 4.5% of the 10.6 ms gap, changing only how to load 3.9%, and both 51.5% (rounded): the choices interact, because MIN caches 4.6

Table 9. The grid: Table 2’s protocol on every card and host we measured, each cell against its own machine’s limit (the exact optimum’s reads on the same trace, the machine’s highest probed host rate, the card’s datasheet rate). *Bound*: what binds the limit; *host* when the optimum’s reads alone take longer than the GPU’s whole read, so the GPU has slack, *both* when the limit runs experts on the CPU until the two paths take equally long, which is where the all-in-VRAM speed can be exceeded. Ours and FreeToken are launch 2 (FreeToken’s backend carried over from host B’s selection); the 40 and 43.75% budgets do not fit a 24 GB card. <sup>†</sup>On the RTX 3090 FreeToken’s carried-over hybrid backend ran below llama.cpp; its own calibration recommends its offload backend there, which we did not run, so its speed is reported and no ratio against it is printed. Mixtral-8x7B (26 GB of experts, top-2 of 8) is ours against llama.cpp only, on the RTX 3090, launch 1.

| Model                                                                                                                                          | Experts on GPU | llama.cpp (tok/s) | FreeToken (tok/s) | Ours (tok/s) | Ours ÷ FreeToken (95% CI) | Ours ÷ llama.cpp     | Limit (tok/s) | Bound | Ours, % of limit |
|------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|-------------------|--------------|---------------------------|----------------------|---------------|-------|------------------|
| <b>RTX 5090</b> , 9950X3D (host B): $B_{\text{host}}$ 88 GB/s, $B_{\text{gpu}}$ 1792 GB/s; tables 0, 0, 1, 1, 2 / 0, 0, 1, 1, 2, 3, 3, 4, 5    |                |                   |                   |              |                           |                      |               |       |                  |
| gpt-oss-120b                                                                                                                                   | 11%            | 34.9              | 54.0              | 69.9         | 1.29 [1.28, 1.31]         | 2.00                 | 172           | host  | 41               |
| gpt-oss-120b                                                                                                                                   | 25%            | 40.0              | 85.6              | 109.2        | 1.27 [1.25, 1.29]         | 2.73                 | 430           | host  | 25               |
| gpt-oss-120b                                                                                                                                   | 40%            | 47.7              | 132.2             | 152.5        | 1.15 [1.13, 1.17]         | 3.20                 | 519           | both  | 29               |
| Qwen3-30B-A3B                                                                                                                                  | 12.5%          | 19.5              | 38.7              | 40.0         | 1.03 [1.02, 1.04]         | 2.05                 | 96            | host  | 42               |
| Qwen3-30B-A3B                                                                                                                                  | 25%            | 22.1              | 54.7              | 63.1         | 1.15 [1.13, 1.17]         | 2.86                 | 214           | host  | 30               |
| Qwen3-30B-A3B                                                                                                                                  | 43.75%         | 28.4              | 103.1             | 108.2        | 1.05 [1.04, 1.06]         | 3.81                 | 308           | both  | 35               |
| <b>RTX 5090</b> , 9950X (host S): $B_{\text{host}}$ 71 GB/s, $B_{\text{gpu}}$ 1792 GB/s; tables 0, 0, 1, 2, 3 / 0, 0, 1, 2, 2, 3, 4, 5, 6      |                |                   |                   |              |                           |                      |               |       |                  |
| gpt-oss-120b                                                                                                                                   | 11%            | 28.9              | 48.0              | 57.9         | 1.21 [1.19, 1.22]         | 2.00                 | 140           | host  | 41               |
| gpt-oss-120b                                                                                                                                   | 25%            | 33.6              | 78.7              | 94.1         | 1.20 [1.16, 1.22]         | 2.80                 | 351           | host  | 27               |
| gpt-oss-120b                                                                                                                                   | 40%            | 40.1              | 122.2             | 133.6        | 1.09 [1.08, 1.11]         | 3.34                 | 514           | both  | 26               |
| Qwen3-30B-A3B                                                                                                                                  | 12.5%          | 16.1              | 32.7              | 33.5         | 1.03 [1.02, 1.04]         | 2.08                 | 78            | host  | 43               |
| Qwen3-30B-A3B                                                                                                                                  | 25%            | 18.5              | 51.4              | 53.9         | 1.05 [1.04, 1.06]         | 2.91                 | 174           | host  | 31               |
| Qwen3-30B-A3B                                                                                                                                  | 43.75%         | 23.8              | 98.3              | 95.8         | 0.97 [0.96, 0.99]         | 4.02                 | 305           | both  | 31               |
| <b>RTX 4090</b> , i5-12400 (job 091): $B_{\text{host}}$ 37 GB/s, $B_{\text{gpu}}$ 1008 GB/s; tables 0, 0, 1, 2, 2 / 0, 0, 1, 1, 2, 3, 4, 4, 5  |                |                   |                   |              |                           |                      |               |       |                  |
| gpt-oss-120b                                                                                                                                   | 11%            | 18.1              | 23.8              | 32.7         | 1.37 [1.35, 1.39]         | 1.81                 | 74            | host  | 44               |
| gpt-oss-120b                                                                                                                                   | 25%            | 21.0              | 42.5              | 55.0         | 1.30 [1.26, 1.33]         | 2.62                 | 184           | host  | 30               |
| Qwen3-30B-A3B                                                                                                                                  | 12.5%          | 10.5              | 18.2              | 18.8         | 1.03 [1.02, 1.05]         | 1.78                 | 41            | host  | 46               |
| Qwen3-30B-A3B                                                                                                                                  | 25%            | 12.3              | 30.0              | 31.3         | 1.05 [1.03, 1.06]         | 2.54                 | 91            | host  | 34               |
| <b>RTX 3090</b> , i9-11900KF (job 092): $B_{\text{host}}$ 45 GB/s, $B_{\text{gpu}}$ 936 GB/s; tables 0, 0, 0, 0, 0 / 0, 0, 0, 0, 0, 0, 0, 0, 0 |                |                   |                   |              |                           |                      |               |       |                  |
| gpt-oss-120b                                                                                                                                   | 11%            | 22.8              | 8.4 <sup>†</sup>  | 37.3         | –                         | 1.64                 | 88            | host  | 42               |
| Qwen3-30B-A3B                                                                                                                                  | 12.5%          | 12.3              | 9.0 <sup>†</sup>  | 19.2         | –                         | 1.56                 | 49            | host  | 39               |
| Qwen3-30B-A3B                                                                                                                                  | 25%            | 13.8              | 11.2 <sup>†</sup> | 30.1         | –                         | 2.18                 | 110           | host  | 27               |
| Mixtral-8x7B Q4_K_M                                                                                                                            | $C=2$ (25%)    | 8.22              | –                 | 8.34         | –                         | 1.015 [1.008, 1.024] | –             | –     | –                |

times as many experts per token as the deployed policy and, loaded by the CPU, each is read twice. We credit each choice with its effect averaged over the two orders, its Shapley value (Fawcett & Hoos, 2016; Jain, 1991) (26.1% and 25.5% here); Table 10 also lists each alone and the interaction. Copying ahead is nested under MIN’s set (12.7%); what no oracle touches is the rest (35.7%). On the 13 hosts whose link reads at least half as fast as the CPU, either change alone closes at most 26% of the gap and both together 14–60%; on the 2 below that line, loading in the step costs time at 3 of 4 host-budgets, and at 3 of 4 on 2 more (job 103). These use the greedy set. With the fewest-admission set (job 104), both together close 6–49% at 11% on its three machines, against –26–39% with the greedy set, and the interaction remains (1–33%).

## F THE ACCOUNTING AS A MODEL

Between the measured time of a cell and its limit lie five modelled fixes: overlap of host reads with GPU work (in Eq. (1)’s form, which also rebalances experts onto the CPU), the optimum’s reads instead of the online policy’s (*foresight*), the best online policy’s reads at the probe’s highest rate instead of the engine’s through Eq. (4) (policy and read paths), the engine’s per-token host work, and GPU kernels at the datasheet rate. Adding them in a fixed order makes the first one added look largest, so we report each fix’s Shapley value, its average marginal cost over every order (Fig. 6), a method with a precedent in performance debugging of microservices (Li et al., 2024). Two of the five terms are definitions rather than measurements: the GPU term is the residual, its factor calibrated per cell so that the model with no fix applied reproduces the measured time, and “overlap” is Eq. (1)’s form. On host B the model has two regimes. In the four host-bound cells missing foresight is the largest cost, 37–53% of the gap (26–31% of the token; in the running example 37%, where it edges overlap by 0.14 ms and is the largest marginal cost in 70 of the 120 orders); in the two GPU-bound cells host reads serialised with GPU work is largest, 39–46% of the gap, and foresight second at 23%. Policy and read paths cost 12–16%, per-token host work 3–10%,

Table 10. An order-free accounting of the gap between the deployed cache and the bound at the host-bound budgets, as percentages of the gap (ms per token). The engine ran both choices of what to cache (the deployed policy’s admissions or MIN’s) with both ways to load a cached expert (two reads: the CPU serves it, then a copy; one read: a copy in the step). *Load alone* and *cache alone* are each choice’s effect with the other at the deployed level; the *interaction* is how much more MIN’s set is worth loaded once than loaded twice. The Shapley values average each choice’s effect over the other’s two levels and add up, with the prefetch (nested under MIN’s set, since it needs foresight) and the rest, to 100%. Panel: mean over its hosts (number in parentheses) and, in small type, the range over its hosts (not an interval); a fast link reads at least half as fast as the host’s CPU, a slow one less. Last rows: the new machines of the follow-up launches, with their link-to-CPU ratio.

| Host                        | Budget      | Gap (ms) | Load alone       | Cache alone   | Interaction   | Shapley value cache | load             | Prefetch      | Rest          |
|-----------------------------|-------------|----------|------------------|---------------|---------------|---------------------|------------------|---------------|---------------|
| O4                          | gpt-oss 11% | 10.6     | 4                | 4             | 43            | 26                  | 25               | 13            | 36            |
| O4                          | gpt-oss 25% | 8.5      | 4                | 22            | 17            | 31                  | 13               | 22            | 34            |
| O4                          | Qwen3 12.5% | 18.4     | 3                | −1            | 58            | 28                  | 32               | 4             | 36            |
| O4                          | Qwen3 25%   | 14.1     | 2                | 14            | 37            | 33                  | 21               | 17            | 30            |
| O5                          | gpt-oss 11% | 11.4     | 0                | −2            | 28            | 12                  | 14               | 17            | 57            |
| O5                          | gpt-oss 25% | 9.1      | −4               | 20            | 10            | 25                  | 1                | 21            | 52            |
| O5                          | Qwen3 12.5% | 19.8     | 1                | −13           | 43            | 9                   | 23               | 7             | 62            |
| O5                          | Qwen3 25%   | 15.3     | 0                | 14            | 17            | 22                  | 9                | 16            | 53            |
| Panel, fast link (8)        | gpt-oss 11% | 11.6     | 1 (−2 to 4)      | 3 (−2 to 6)   | 38 (30 to 47) | 22 (14 to 26)       | 20 (14 to 27)    | 13 (−4 to 20) | 45 (38 to 55) |
| Panel, fast link (8)        | gpt-oss 25% | 9.0      | 0 (−4 to 3)      | 22 (18 to 26) | 15 (10 to 19) | 29 (25 to 35)       | 8 (1 to 12)      | 20 (9 to 23)  | 43 (38 to 49) |
| Panel, slow link (2)        | gpt-oss 11% | 9.7      | −6 (−9 to −3)    | −4 (−9 to 0)  | 3 (−6 to 11)  | −3 (−11 to 6)       | −5 (−12 to 2)    | 17 (9 to 24)  | 91 (83 to 99) |
| Panel, slow link (2)        | gpt-oss 25% | 7.8      | −17 (−22 to −11) | 19 (16 to 23) | 3 (3 to 3)    | 21 (17 to 25)       | −15 (−21 to −10) | 22 (17 to 26) | 73 (68 to 78) |
| 5700X3D (0.92)              | gpt-oss 11% | 14.9     | 4                | −2            | 55            | 25                  | 31               | 8             | 36            |
| 5700X3D (0.92)              | gpt-oss 25% | 11.5     | 4                | 22            | 24            | 34                  | 17               | 18            | 31            |
| 9950X, x8 (0.54)            | gpt-oss 11% | 8.3      | 0                | −3            | 17            | 5                   | 8                | 28            | 59            |
| 9950X, x8 (0.54)            | gpt-oss 25% | 7.4      | −15              | 13            | 17            | 22                  | −6               | 26            | 59            |
| 9800X3D, slower link (0.61) | gpt-oss 11% | 10.8     | −2               | −4            | 29            | 11                  | 12               | 20            | 57            |
| 9800X3D, slower link (0.61) | gpt-oss 25% | 8.9      | −5               | 20            | 12            | 26                  | 1                | 20            | 54            |

and GPU kernels below datasheet, net of the overlap the engine achieves, 4–16%. The regimes are a property of the host: recomputed for the three oracle hosts, whose host memory is weaker, foresight edges overlap at three of their six GPU-bound cells. The model prices foresight and overlap as nearly independent players, and the interior states it values are not realised by any online policy. The rest of this section measures them.

## G FORESIGHT IN THE ENGINE

Four jobs ran oracle policies in the engine, each on a rented RTX 5090 host with weaker host memory than host B’s. Jobs 093–097 report ratios of mean speeds rather than of mean times per token; recomputed either way, no ratio moves by more than 0.006. The fourth (096) is the confirmatory factorial of Fig. 1: ten configurations per cell, each in a fresh context, in an order shuffled per cell with a recorded seed, on hosts O4 (Ryzen 9 9950X: 43, 45 and 51 GB/s by the CPU, over the link and at best) and O5 (Ryzen 9 9950X3D: 47, 27 and 51 GB/s), with ten predictions per host committed before either started (prereg/foresight\_outcome\_096.md). They held at most cells. The failures: fetch’s gain above the predicted band at Qwen3 25% on O4 and different between the hosts by up to 0.26 against the 0.15 predicted; Belady’s single-read prefetch faster than predicted at Qwen3 12.5% on O4, with reads below the predicted 1.25–1.9× the MIN prefetch’s at Qwen3 12.5% on both hosts and at gpt-oss 11% on O4 (down to 1.09×); the bytes share at Qwen3 12.5% on O4 (60% of the gap against a predicted 25–55%); and the reads fetch-on-admit saves at gpt-oss 40% on O5, just above the predicted 22%. Reading the online policy’s admissions once came to 0.97–1.03× at the host-bound cells, at the bottom of the predicted 0.97–1.12× although its reads fell as predicted, and to 0.95–1.02× at the GPU-bound cells. Table 11 lists every cell of the single-read factorial of Section 4 on host O3 (a Ryzen 9 9950X whose probe reads 53 GB/s by the CPU at the helper count, 43 over the link, 60 with both and 61.5 at best; the model’s FETCH tables 0, 0, 1, 2, 2 for gpt-oss and 0, 0, 1, 1, 2, 3, 4, 4, 5 for Qwen3), and Table 12 the milliseconds per token of the engine states the three jobs realised beside each host’s limit and the model’s values. Table 13 lists the first job (093), the hit-optimal oracle with a window sweep on host O1 (a Ryzen 9 9950X: 44, 46, 49 and 50.5 GB/s; tables 0, 0, 1, 2, 3 and 0, 0, 1, 2, 3, 4, 5, 6, 7), and Table 14 the second (094), MIN with bypass with background admissions beside the hit-optimal oracle on host O2 (a Ryzen 9 7950X: 44, 46, 50 and 50.6 GB/s; the same tables). On O1 and O2 every cell but gpt-oss 40% and Qwen3 43.75% is host-bound at the limit; on O3, whose memory is stronger, the same four.

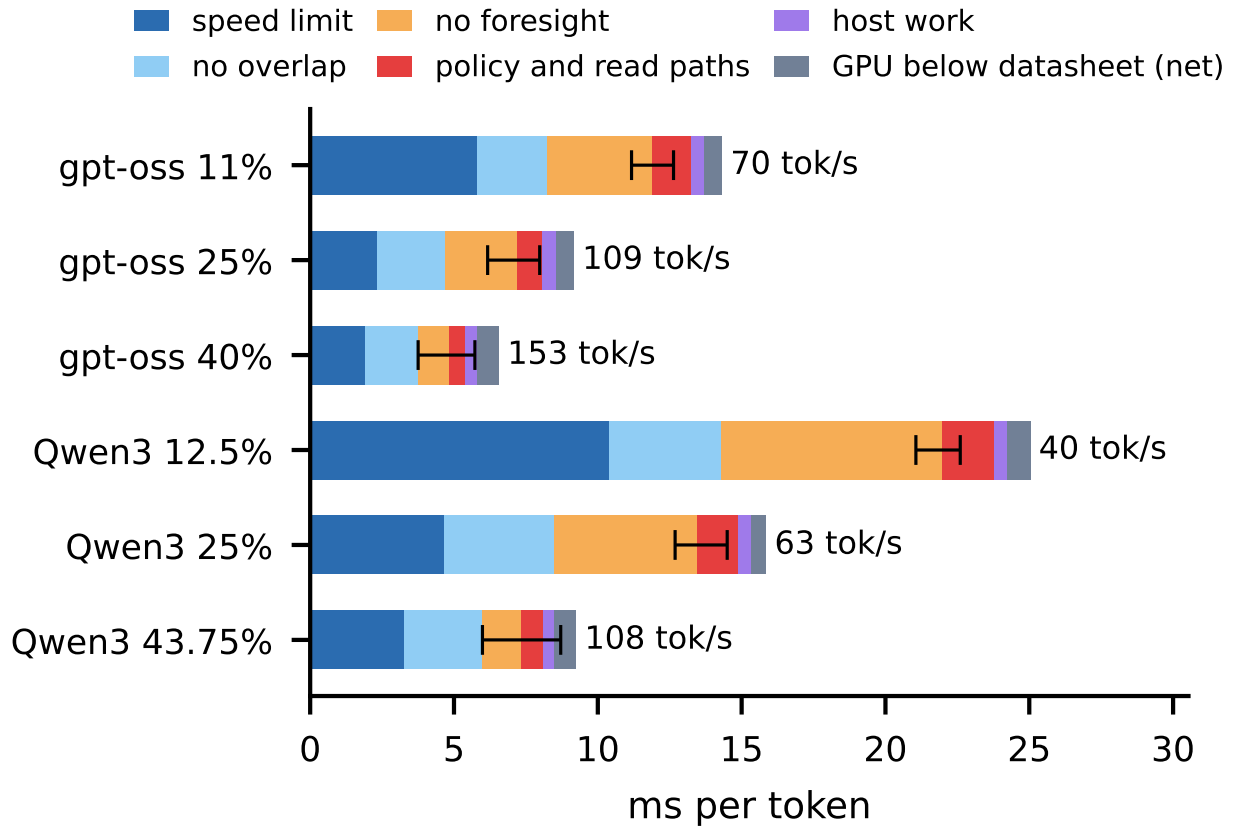


Figure 6. The accounting as a model: Shapley attribution of the gap between our cache’s measured time per token and the bound, for every cell of Table 2 on host B, averaged over all 120 orders in which the five fixes can be applied. Fixes: overlapping host reads with GPU work, which in Eq. (1)’s form also moves experts onto the CPU until the two paths balance; the optimum’s reads instead of the best online policy’s (foresight); the best online policy’s reads at the probe’s highest rate instead of the engine’s reads through Eq. (4) (policy and read paths); the engine’s per-token host work; and GPU kernels at the datasheet rate, the term that closes the model on the measured time. Whiskers: the range of foresight’s marginal cost over the orders.

**The single-read oracles.** In the fetch oracle the step’s plan is computed before the layer runs from the trace’s record of the current step: the misses MIN with bypass would admit are fetched by the engine’s FETCH operation into their victims’ slots and run from there, and every other miss runs on the CPU; the oracle refuses an admission when no resident’s next use is further than the candidate’s. The single-read prefetch admits, each step, the non-resident experts whose first use is at least  $d$  steps away ( $d=2$  unpaced, 3 paced), provided the victim’s next use is further than the candidate’s use after that first one; an expert used sooner than  $d$  steps away misses on the CPU, or is fetched when the two are combined. The same policies simulated on the trace with the engine’s semantics (the lookahead within the sequence, residents carried over with their next use recomputed at each sequence start) predict the engine’s reads and hit rates: the fetch oracle’s hit rate within 0.4 points and its reads within 0 to 4%, the combined oracle’s hits within two points (Table 11). The simulator the job’s predictions were registered against gave a resident carried over from the previous sequence no next use, so it was evicted first; that over-counted the fetch oracle’s reads by 17% at the two GPU-bound cells, which is why two of its reads clauses failed (Appendix B). Each configuration of llama-ec-bench starts with a cold cache in a fresh context. The paced copy path copies one 16 MB piece at a time on a single copier thread and publishes a landed copy at the start of the next step; the unpaced path copies each admission as one transfer. The hit-optimal oracle’s background admissions drive host memory at 71–82% of the probe’s best rate at the host-bound cells, the prefetching single-read oracles at 64–85% (the paced

Table 11. The single-read oracles (job 095; RTX 5090, Ryzen 9 9950X, host memory 61.5 GB/s at best) beside the hit-optimal one. Per cell: the online policy’s speed and its share of this host’s limit; for each oracle with the whole sequence in view, its ratio to the online policy (paired, 95% interval), hit rate, host reads per token (CPU misses + fetches + admissions) as a multiple of the optimum’s (each read counted once), share of the limit and the calibrated model on its own counters against its measured time. *fetch*: MIN with bypass, admitted misses fetched into their slot this step; *lead*: the scheduled single-read prefetch (first use at least two steps away); *both*: fetch and lead; *paced*: both on the paced copy path with a three-step lead; *hit-opt.*: the hit-optimal oracle (Belady prefetch without bypass, unpaced). The simulation columns give the trace’s prediction of reads and hits for fetch and both.

| Cell         | online |        | ratio             | fetch |          |        | model                  | both  |      |          | hit-opt.          |       |      | lead     |        | paced |        | simulated |        |             |      |            |
|--------------|--------|--------|-------------------|-------|----------|--------|------------------------|-------|------|----------|-------------------|-------|------|----------|--------|-------|--------|-----------|--------|-------------|------|------------|
|              | tok/s  | % lim. |                   | hits  | reads/R* | % lim. |                        | ratio | hits | reads/R* | % lim.            | ratio | hits | reads/R* | % lim. | ratio | % lim. | ratio     | % lim. | fetch reads | hits | both reads |
| gpt-oss 11%  | 55.5   | 46     | 1.29 [1.28, 1.29] | 72    | 1.04     | 59     | -8% 1.39 [1.39, 1.40]  | 86    | 1.23 | 64       | 0.91 [0.90, 0.92] | 98    | 1.77 | 42       | 1.26   | 58    | 1.45   | 67        | 39     | 73          | 46   | 87         |
| gpt-oss 25%  | 90.3   | 30     | 1.32 [1.31, 1.33] | 89    | 1.08     | 39     | -13% 1.49 [1.48, 1.50] | 97    | 1.45 | 44       | 1.14 [1.12, 1.15] | 99    | 2.11 | 34       | 1.45   | 43    | 1.66   | 50        | 16     | 89          | 22   | 99         |
| gpt-oss 40%  | 128.8  | 25     | 1.25 [1.24, 1.27] | 95    | 1.14     | 32     | -14% 1.39 [1.37, 1.41] | 99    | 1.60 | 35       | 1.27 [1.25, 1.28] | 99    | 2.20 | 32       | 1.38   | 35    | 1.49   | 38        | 7      | 95          | 12   | 100        |
| Qwen3 12.5%  | 32.3   | 48     | 1.34 [1.34, 1.35] | 74    | 1.04     | 64     | -9% 1.42 [1.41, 1.43]  | 84    | 1.20 | 68       | 0.69 [0.68, 0.71] | 77    | 2.47 | 33       | 1.22   | 58    | 1.41   | 68        | 100    | 74          | 115  | 84         |
| Qwen3 25%    | 51.8   | 34     | 1.38 [1.37, 1.39] | 88    | 1.04     | 48     | -15% 1.48 [1.46, 1.49] | 96    | 1.42 | 51       | 1.06 [1.05, 1.08] | 99    | 2.00 | 37       | 1.41   | 48    | 1.67   | 57        | 45     | 88          | 61   | 97         |
| Qwen3 43.75% | 91.4   | 30     | 1.26 [1.25, 1.28] | 96    | 1.08     | 38     | -17% 1.41 [1.39, 1.42] | 99    | 1.62 | 42       | 1.27 [1.26, 1.29] | 100   | 2.29 | 38       | 1.40   | 42    | 1.50   | 45        | 15     | 96          | 23   | 100        |

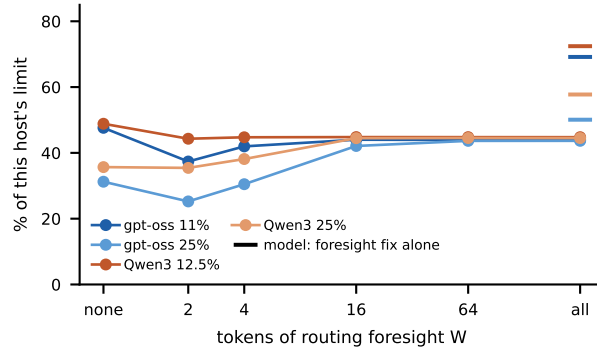


Figure 7. The first oracle job (093, host O1): speed as a percentage of this host’s limit against the window  $W$  of future routing the hit-optimal oracle sees, at the four host-bound cells. “none” is the online policy; “all” the whole sequence; points are connected for the trend only. The tick at the right is the model’s foresight-only term for the cell, the speed it predicts with that one fix; the oracle’s host traffic sits at 89–94% of the probe’s best rate at every window, so its speed is  $R^*$  over its reads times that utilisation, 44–45% of the limit with the whole sequence in view.

one 71–82%), the online policy itself at 66–82%, the fetch oracle at 42–67%. The predictions and their outcomes are in `prereg/foresight_outcome_095.md`: of 56 clauses, 44 held and 12 failed, all of size (Appendix B).

**The double-reading oracles.** Both read the routing of the next  $W$  decode steps; their copies ride the paced admission path, and an expert still in flight at its step runs on the CPU like any other miss. The *hit-optimal* oracle admits the experts those steps will route to, soonest first, evicting the resident whose next use is furthest (Belady within the window), for one future use as readily as for ten. The *bytes-optimal* oracle is MIN with bypass with its admissions made through the background path: a miss is served where it is, by the CPU or a demand copy, and admitted only if its next use comes before the furthest next use among the residents. With the whole sequence in view the hit-optimal oracle runs  $1.40 \times [1.38, 1.41]$  the online policy at gpt-oss 25% and  $1.25 \times [1.24, 1.26]$  at Qwen3 25% on host O1,  $1.63 \times$  and  $1.41 \times$  on O2, and  $1.48$ – $1.62 \times$  at the two GPU-bound cells; half of its gain at gpt-oss 25% arrives between  $W=4$  and 16 (9.2 tokens by log interpolation; Qwen3 25%: 6.1 against the trace’s 3.9). Where the host binds hardest it loses, 3–8% at gpt-oss 11% and Qwen3 12.5% on both hosts, and at every window on O1 ( $0.78$ – $0.92 \times$ ): it reads more host bytes than the online policy (82 and 195 experts per token against 63 and 160),  $2.02$ – $2.18 \times$  the optimum’s  $R^*$ , because Belady admits an expert used once where the optimum bypasses it and re-copies what it evicts (its admissions, each one read, are most of its reads; a miss served on the CPU before its copy lands is the second read of that expert). An unpaced copy path, run at the four host-bound cells of O1, reaches 98–99% hits at three of them and still loses at gpt-oss 11% and Qwen3 12.5% (where it hits 75%), so copies that arrive in time do not remove the loss there: the extra bytes cause it. The bytes-optimal oracle gains 13–23% at the four cells of 25% and above and 0–3% at the two below (6% with a 16-token window); its misses fall (30 to 20 and 85 to 64 per token at 25%) but its total reads, misses plus the admissions that read each admitted expert again, are  $1.73$ – $2.04 \times R^*$  and exceed the online policy’s by 4–5% at the two lowest budgets; the model on its own counters predicts its time within 9% at

Table 12. The accounting, measured: milliseconds per token of the engine’s states that the oracle jobs realised, on each oracle host, beside that host’s limit and the model’s values for the states it prices; O1–O3 ran jobs 093–095 in a fixed order, O4 and O5 job 096 in a shuffled order. *online*: the deployed policy; *foa*: the same with each admission fetched into its slot in the step that misses it (one read, no foresight); *no ovl.*: its within-layer overlap off; *all CPU*: no FETCH table; *bypass*: MIN with bypass with background admissions (two reads per admitted expert); *hit-opt.*: Belady prefetch without bypass, which admits what the optimum bypasses, unpaced, and the same on the paced copy path (the first *paced* column; O1 and O2 ran only that form); *nb2*: Belady within a two-step lead with each admission read once; *fetch*: MIN with bypass, admitted misses fetched into their slot (one read, serialised: the model’s foresight-only state  $v(F)$ ); *both*: fetch and the scheduled single-read prefetch (one read, overlapped: the model’s  $v(O, F)$ ); the second *paced*: both on the paced copy path with a three-step lead; *limit*: the limit on that host;  $v(F)$ ,  $v(O, F)$ ,  $v(all)$ : the model.

| Host | Cell         | online | foa   | no ovl. | all CPU | bypass | hit-opt. | paced | nb2   | fetch | both  | paced | limit | $v(F)$ | $v(O, F)$ | $v(all)$ |
|------|--------------|--------|-------|---------|---------|--------|----------|-------|-------|-------|-------|-------|-------|--------|-----------|----------|
| O1   | gpt-oss 11%  | 21.12  | –     | 21.01   | 22.91   | –      | –        | 22.88 | –     | –     | –     | –     | 10.06 | 14.54  | 11.27     | 10.06    |
| O1   | gpt-oss 25%  | 12.89  | –     | 12.91   | 14.21   | –      | –        | 9.22  | –     | –     | –     | –     | 4.03  | 8.04   | 5.10      | 4.03     |
| O1   | gpt-oss 40%  | 8.90   | –     | 8.95    | –       | –      | –        | 5.99  | –     | –     | –     | –     | 1.97  | 5.75   | 3.84      | 1.97     |
| O1   | Qwen3 12.5%  | 36.92  | –     | 36.91   | 43.29   | –      | –        | 40.29 | –     | –     | –     | –     | 18.04 | 24.90  | 19.29     | 18.04    |
| O1   | Qwen3 25%    | 22.73  | –     | 22.76   | 26.60   | –      | –        | 18.22 | –     | –     | –     | –     | 8.11  | 14.04  | 9.22      | 8.11     |
| O1   | Qwen3 43.75% | 12.71  | –     | 12.67   | –       | –      | –        | 8.12  | –     | –     | –     | –     | 3.32  | 8.25   | 5.51      | 3.32     |
| O2   | gpt-oss 11%  | 20.85  | –     | –       | –       | 20.17  | –        | 21.57 | –     | –     | –     | –     | 10.04 | 14.27  | 11.96     | 10.04    |
| O2   | gpt-oss 25%  | 12.72  | –     | –       | –       | 10.77  | –        | 7.80  | –     | –     | –     | –     | 4.02  | 7.86   | 5.86      | 4.02     |
| O2   | gpt-oss 40%  | 8.79   | –     | –       | –       | 7.13   | –        | 5.62  | –     | –     | –     | –     | 1.97  | 5.63   | 3.78      | 1.97     |
| O2   | Qwen3 12.5%  | 36.64  | –     | –       | –       | 36.58  | –        | 38.74 | –     | –     | –     | –     | 18.00 | 24.70  | 20.06     | 18.00    |
| O2   | Qwen3 25%    | 22.24  | –     | –       | –       | 19.67  | –        | 15.80 | –     | –     | –     | –     | 8.09  | 13.58  | 9.90      | 8.09     |
| O2   | Qwen3 43.75% | 12.02  | –     | –       | –       | 10.11  | –        | 7.42  | –     | –     | –     | –     | 3.32  | 7.56   | 4.90      | 3.32     |
| O3   | gpt-oss 11%  | 18.01  | –     | –       | –       | –      | 19.79    | –     | –     | 14.00 | 12.91 | 12.39 | 8.26  | 12.63  | 9.49      | 8.26     |
| O3   | gpt-oss 25%  | 11.08  | –     | –       | –       | –      | 9.74     | –     | –     | 8.42  | 7.44  | 6.66  | 3.31  | 7.05   | 4.28      | 3.31     |
| O3   | gpt-oss 40%  | 7.77   | –     | –       | –       | –      | 6.14     | –     | –     | 6.20  | 5.58  | 5.20  | 1.95  | 5.24   | 3.56      | 1.95     |
| O3   | Qwen3 12.5%  | 30.95  | –     | –       | –       | –      | 44.54    | –     | –     | 23.02 | 21.78 | 21.89 | 14.81 | 20.35  | 16.06     | 14.81    |
| O3   | Qwen3 25%    | 19.32  | –     | –       | –       | –      | 18.17    | –     | –     | 14.01 | 13.09 | 11.58 | 6.66  | 11.68  | 7.76      | 6.66     |
| O3   | Qwen3 43.75% | 10.94  | –     | –       | –       | –      | 8.59     | –     | –     | 8.66  | 7.78  | 7.28  | 3.30  | 7.24   | 4.76      | 3.30     |
| O4   | gpt-oss 11%  | 20.59  | 20.18 | –       | –       | 20.11  | 19.78    | 22.03 | 19.64 | 15.13 | 14.06 | 13.78 | 10.02 | 13.77  | 11.63     | 10.02    |
| O4   | gpt-oss 25%  | 12.43  | 12.07 | –       | –       | 10.55  | 9.76     | 8.13  | 9.61  | 8.72  | 7.65  | 6.87  | 4.01  | 7.40   | 5.34      | 4.01     |
| O4   | gpt-oss 40%  | 8.39   | 8.19  | –       | –       | 6.91   | 6.15     | 5.38  | 6.03  | 6.28  | 5.59  | 5.21  | 1.97  | 5.12   | 3.20      | 1.97     |
| O4   | Qwen3 12.5%  | 36.36  | 35.83 | –       | –       | 36.54  | 48.80    | 39.66 | 31.64 | 25.29 | 24.01 | 24.58 | 17.97 | 23.98  | 19.88     | 17.97    |
| O4   | Qwen3 25%    | 22.09  | 21.75 | –       | –       | 20.06  | 17.74    | 16.93 | 17.57 | 14.60 | 13.50 | 12.24 | 8.08  | 13.12  | 9.54      | 8.08     |
| O4   | Qwen3 43.75% | 11.94  | 11.74 | –       | –       | 10.09  | 8.44     | 7.17  | 8.33  | 8.69  | 7.72  | 7.29  | 3.32  | 7.32   | 4.56      | 3.32     |
| O5   | gpt-oss 11%  | 21.26  | 21.31 | –       | –       | 21.47  | 33.34    | 25.57 | 33.03 | 18.34 | 17.97 | 16.39 | 9.94  | 15.22  | 11.20     | 9.94     |
| O5   | gpt-oss 25%  | 13.01  | 13.35 | –       | –       | 11.18  | 15.87    | 11.17 | 15.60 | 10.60 | 10.54 | 8.68  | 3.98  | 9.13   | 5.06      | 3.98     |
| O5   | gpt-oss 40%  | 8.73   | 9.22  | –       | –       | 7.16   | 8.34     | 6.25  | 8.13  | 7.27  | 6.87  | 5.91  | 1.97  | 6.26   | 3.62      | 1.97     |
| O5   | Qwen3 12.5%  | 37.63  | 37.42 | –       | –       | 40.11  | 69.42    | 44.89 | 49.11 | 31.37 | 31.27 | 30.01 | 17.82 | 27.95  | 19.27     | 17.82    |
| O5   | Qwen3 25%    | 23.24  | 23.27 | –       | –       | 21.12  | 30.78    | 21.76 | 30.47 | 18.51 | 19.64 | 16.03 | 8.01  | 16.77  | 9.15      | 8.01     |
| O5   | Qwen3 43.75% | 12.49  | 12.92 | –       | –       | 10.32  | 12.08    | 9.00  | 11.86 | 10.21 | 9.61  | 8.23  | 3.31  | 8.94   | 5.02      | 3.31     |

every cell, and the bytes it saves account for 4–21 points of its 13–23% gain at the four cells (most of it at gpt-oss 40% and Qwen3 43.75%, a third at Qwen3 25%). The hit-optimal oracle reads  $1.01\text{--}1.13\times$  the bytes-optimal one’s host bytes on O2 and gains 41–63% at the same four cells, and the model, with its GPU term calibrated on the online run, over-predicts those runs’ time per token by 37–53% (on O1, 24–39%): its gain is the overlap of copies issued ahead of their use with the step, which the model charges in sequence. Pacing the online policy’s own copies changes it by under 1% ( $0.997\text{--}1.005\times$ ). With no FETCH table (every miss on the CPU) the online policy runs 8–15% slower at the four host-bound cells; with the engine’s within-layer overlap of a layer’s CPU misses with its GPU hits turned off it changes by under 1% at every cell (0.5% faster to 0.6% slower), so the overlap the accounting charges is across the step and in rebalancing, not within a layer. The bytes-optimal oracle with a 16-token window keeps 34–176% of the whole-sequence gain at the five cells where that gain’s interval excludes zero (more than all of it at gpt-oss 11%, where the shorter window admits less).

**Why the bytes-optimal oracle’s misses stay above the optimum’s reads.** Besides the second read, its misses alone are  $1.31\text{--}1.47\times$  the optimum’s reads at the four host-bound cells (at the two GPU-bound ones the ideal simulation below is itself 6–10% above the optimum, which sees across sequences). The engine’s paced copy path moves one 16 MB piece at a time from pageable host memory on a single copier thread, behind the demand copies, publishes a landed copy at the start of the next step, and admits nothing while 64 copies are queued; at the 16–30 admissions per token (210–285 MB) of the two lowest budgets it lags the admissions by steps, and an expert still in flight is a miss the CPU reads again. The last four columns of Table 14 simulate MIN with bypass on the measured routing trace with the engine’s own semantics (the lookahead within the sequence, residents carried over, no queue cap) and a copy that lands  $d$  steps after it is issued, during which the expert misses and is not re-admitted: at  $d=1$ , the ideal, the misses come within 0–4% of the limit’s optimum at the host-bound cells; the engine’s misses meet the simulation at  $d=2.8\text{--}4.7$  there and at 3.0–3.5 at the GPU-bound cells, three to five steps. At the fitted latency the simulation admits  $1.1\text{--}1.9\times$  what the engine admitted, so the queue cap, not only the latency, keeps

Table 13. The hit-optimal oracle in the engine (job 093; RTX 5090, Ryzen 9 9950X, host memory 50.5 GB/s at best). tok/s of the online policy and of the oracle with the whole sequence in view; their ratio, paired by sequence, with its 95% interval; hit rate and misses per token of each (experts read from host memory by the CPU or by a demand copy), the optimum’s reads in parentheses, and the oracle’s background admissions (Belady prefetch without bypass: it admits what the optimum bypasses); speed as a percentage of this host’s limit; the accounting’s foresight-only term  $v(F)$  for the cell in tok/s; the share of that term the oracle recovered and the share of the gap to the limit it closed.

| Cell         | online | oracle | ratio             | hits % (opt.) | misses/token (opt.) | oracle adm. | % of limit  | $v(F)$ | recovered % | closed % |
|--------------|--------|--------|-------------------|---------------|---------------------|-------------|-------------|--------|-------------|----------|
| gpt-oss 11%  | 47.4   | 43.7   | 0.92 [0.91, 0.93] | 58 → 67 (73)  | 60 → 47 (38)        | 35          | 47.6 → 44.0 | 68.8   | −27         | −16      |
| gpt-oss 25%  | 77.6   | 108.5  | 1.40 [1.38, 1.41] | 79 → 92 (89)  | 30 → 11 (15)        | 20          | 31.2 → 43.7 | 124.4  | 76          | 41       |
| gpt-oss 40%  | 112.4  | 166.9  | 1.48 [1.46, 1.51] | 89 → 98 (95)  | 15 → 3 (7)          | 12          | 22.1 → 32.8 | 174.0  | 92          | 42       |
| Qwen3 12.5%  | 27.1   | 24.8   | 0.92 [0.91, 0.92] | 60 → 66 (75)  | 155 → 132 (97)      | 63          | 48.9 → 44.8 | 40.2   | −28         | −18      |
| Qwen3 25%    | 44.0   | 54.9   | 1.25 [1.24, 1.26] | 78 → 89 (89)  | 85 → 42 (43)        | 46          | 35.7 → 44.5 | 71.2   | 52          | 31       |
| Qwen3 43.75% | 78.7   | 123.1  | 1.56 [1.53, 1.59] | 92 → 99 (96)  | 32 → 5 (14)         | 25          | 26.1 → 40.8 | 121.3  | 103         | 49       |

Table 14. The bytes-optimal oracle (MIN with bypass) beside the hit-optimal one on the second oracle host (RTX 5090, Ryzen 9 7950X, host memory 50.6 GB/s at best). Per cell: the online policy’s speed; each oracle’s ratio to it with the whole sequence in view (paired, 95% interval), its hit rate, its misses per token (experts read from host memory by the CPU or by a demand copy; its background admissions in parentheses: the bypass oracle’s are this step’s CPU-served misses, each read a second time by the copy, the hit-optimal one’s are prefetches of what the optimum would bypass as readily as of what it would admit) and the calibrated model on its own counters against its measured time; the optimum’s reads and hit rate from the trace; and MIN with bypass simulated on the trace with the engine’s lookahead (within the sequence) under a copy latency of 1 (ideal), 2 and 3 steps, with the latency at which the simulation meets the engine’s misses.

| Cell         | online | ratio             | bypass oracle, $W=all$ |               |       | ratio             | hit-optimal oracle, $W=all$ |               |       | optimum |      | simulated misses/token |     |     |     |
|--------------|--------|-------------------|------------------------|---------------|-------|-------------------|-----------------------------|---------------|-------|---------|------|------------------------|-----|-----|-----|
|              |        |                   | hits                   | misses (adm.) | model |                   | hits                        | misses (adm.) | model | reads   | hits | $d=1$                  | 2   | 3   | fit |
| gpt-oss 11%  | 48.0   | 1.03 [1.03, 1.04] | 65                     | 50 (16)       | +7%   | 0.97 [0.96, 0.98] | 71                          | 42 (39)       | +19%  | 38.3    | 73   | 39                     | 43  | 47  | 3.9 |
| gpt-oss 25%  | 78.6   | 1.18 [1.18, 1.19] | 86                     | 20 (9)        | +6%   | 1.63 [1.61, 1.66] | 95                          | 7 (23)        | +53%  | 15.3    | 89   | 16                     | 17  | 19  | 4.7 |
| gpt-oss 40%  | 113.8  | 1.23 [1.22, 1.24] | 94                     | 9 (4)         | +2%   | 1.56 [1.54, 1.59] | 98                          | 3 (12)        | +39%  | 6.8     | 95   | 7                      | 8   | 9   | 3.5 |
| Qwen3 12.5%  | 27.3   | 1.00 [1.00, 1.01] | 64                     | 138 (30)      | +5%   | 0.95 [0.94, 0.95] | 67                          | 128 (65)      | +11%  | 96.5    | 75   | 97                     | 122 | 143 | 2.8 |
| Qwen3 25%    | 45.0   | 1.13 [1.12, 1.14] | 83                     | 64 (21)       | +9%   | 1.41 [1.39, 1.43] | 91                          | 34 (52)       | +37%  | 43.4    | 89   | 44                     | 54  | 61  | 3.4 |
| Qwen3 43.75% | 83.2   | 1.19 [1.18, 1.20] | 95                     | 19 (9)        | +1%   | 1.62 [1.58, 1.65] | 99                          | 3 (27)        | +47%  | 13.7    | 96   | 14                     | 17  | 19  | 3.0 |

the engine’s misses above the ideal. The predictions and their outcomes are in `prereg/foresight_outcome_093.md` and `094.md`: of 36 clauses for the first job, 9 held, 2 held on the point estimate, 21 failed and 4 were untested; of 38 for the second, 12 held, 1 on the point estimate, 24 failed and 1 was untested (Appendix B). The second job’s failures are of size, not direction: the bypass oracle’s misses were predicted within 15% of the optimum’s reads and came 29–47% above (the copy latency and the queue cap); its hit rate was predicted within 4 points of the optimum’s and fell 5–11 points short at three cells; its gain was predicted at 10–45% at the host-bound cells and came below that at the two lowest budgets; the model was predicted to over-predict it by 10–40% at the host-bound cells and over-predicted by 5–9%; and the two oracles were predicted within 15% of each other at 25% and above, where the hit-optimal one was  $1.24\text{--}1.38\times$  faster.

**Where a token’s time goes at a GPU-bound budget.** On Qwen3 at 43.75% Nsight Systems splits a token of our cache (FreeToken in parentheses) into 4.93 (5.20) ms of expert copies over PCIe, 2.29 (2.21) of resident-expert products, 1.65 (1.34) of attention projections, 0.49 (0.37) of attention, 0.38 of output head and 0.97 (0.40) of norms, small kernels and cache control, with neither system overlapping its copies with GPU compute. Turning off the within-layer overlap of a layer’s CPU misses with its GPU hits changes speed by under 1% at every budget, and predicting the next layer’s experts (Hwang et al., 2024; Zhu et al., 2025) never lowers host bytes.

## H THE TRACE STUDY IN DETAIL

**Policies.** Table 15 scores LRU, LFU, decayed frequency, ARC, S3-FIFO and a profiled static cache against MIN and against Belady within a window of  $W$  future tokens, at three budgets per model, on each model’s own sampled text (9 traces, the cache carried across conversations). Without admitting every miss, the online policies are close to one another and far from MIN: the best reads 35–110% more, decayed frequency is best at 12 of 27 budget points and S3-FIFO at 10. Where the cache turns over every token ( $C/k \leq 1.5$ ) the six are within 4% of one another; from  $C/k=2$  LRU reads 2–11% more than decayed frequency on eight of nine models, and classic LFU fails at large budgets. Adding *admit every miss* (every miss copied into the slot of the lowest-scored resident not requested now) changes the ranking: it is best at 18 of 27 points, ties at 5, and loses by at most 2.7%; it saves up to 5.8% over the previous best, and the best online policy still reads

Table 15. Host reads per token relative to the exact optimum (Belady’s MIN with bypass over the whole future), median over nine models at three budgets, and the range over all 27 model–budget cells. Event-atomic replay of each model’s own sampled text; every online policy is scored on the last 90% of the trace. Mixtral (8 experts) uses  $C = 2, 3$  and 4.

| Policy                   | $C=E/8$ | $E/4$ | $3E/8$ | range      |
|--------------------------|---------|-------|--------|------------|
| LRU                      | 1.44    | 1.77  | 2.03   | 1.37–2.26  |
| LFU                      | 1.44    | 1.93  | 2.40   | 1.37–6.15  |
| Decayed frequency        | 1.45    | 1.70  | 1.94   | 1.35–2.10  |
| with hysteresis $\kappa$ | 1.46    | 1.75  | 2.00   | 1.36–2.55  |
| ARC                      | 1.44    | 1.76  | 1.99   | 1.37–2.23  |
| S3-FIFO                  | 1.44    | 1.69  | 1.94   | 1.38–2.11  |
| Static, profiled         | 1.92    | 2.74  | 3.48   | 1.43–13.24 |
| Belady, $W=1$            | 1.18    | 1.45  | 1.69   | 1.07–2.06  |
| Belady, $W=4$            | 1.01    | 1.13  | 1.31   | 1.00–1.95  |
| Belady, $W=16$           | 1.00    | 1.00  | 1.01   | 1.00–1.66  |

35–110% more than MIN (prereg/policy\_aa.json). At the small budgets almost every resident turns over within a few tokens, and the expert just requested is a good bet. In our engine admitting every miss reads all of its misses over the link, so its value in time depends on the link (Section 5). A static cache profiled on other text reads  $1.4\text{--}13.2\times$  MIN because the hot set moves: the  $C$  most requested experts of a layer change by 22–58% between windows of 100 tokens and 33–75% between windows of 1,000.

**The horizon, before and after the stronger baseline.** Against decayed frequency with each admission fetched once (the study’s original baseline), MIN reads 26–56% fewer experts (median 41%), and the window that closes half of the gap fitted  $W_{50} \approx 0.59 (C/k)^{1.33}$ ; admitting every miss lowers the baseline at 23 of 26 points by up to 6.7%, raises  $W_{50}$  by a median 9% (at most 42%), and refits it to  $0.67 (C/k)^{1.31}$ , with a leave-one-model-out exponent range of 1.25–1.36 and, without the points below one token, 1.35 (it was 1.51 against the original baseline). Ninety percent of the gap needs a median of  $2.7 C/k$  tokens.

**A learned admission order.** We trained a logistic model of whether an expert is requested within the next  $C/k$  tokens on features the engine has when it plans a step (decayed request counts at five half-lives, the time since the last request, whether the expert is requested now, and transition scores from the previous token’s experts at the same layer and, through an 8-bit table, at every layer), fitted on each model’s mixed-domain trace and replayed on the AIME routing, so no test text is seen in training (Table 16). With each admission read once it reads 6–12% fewer experts than decayed frequency with the same single-read admission at the four host-bound budgets, closing 17–30% of that policy’s excess over MIN; admitting every miss alone closes most of that at Qwen3 12.5% and little at gpt-oss (prereg/learned.offline.json and the research checks). In the engine (job 097, predictions committed before launch) it reads 6–13% fewer experts than the single-read online policy at the host-bound budgets on both hosts, but runs  $1.01\text{--}1.03\times$  it on O4’s machine and  $0.90\text{--}0.94\times$  on host O6: the reads it saves are CPU reads, bought with more fetches in the step, each a copy its layer waits on, and with 237–585  $\mu\text{s}$  of host time per token (over all six budgets).

Table 16. A learned admission order, offline: host reads per token on the AIME-25 routing the engine runs, of the decayed frequency with each admission read once (single read; within 10% of the engine’s fetch-on-admit reads on hosts O4 and O5), the learned order with single-read admission (trained on the model’s mixed-domain trace; 8-bit cross-layer table, as in the engine), Belady within one and four tokens of perfect foresight, and MIN with bypass in its no-evict form (Table 18). *closed*: the share of the single-read policy’s excess over MIN that the learned order removes. <sup>†</sup>host-bound.

| cell                     | single read | learned | $W=1$ | $W=4$ | MIN   | closed |
|--------------------------|-------------|---------|-------|-------|-------|--------|
| gpt-oss 11% <sup>†</sup> | 59.4        | 56.0    | 54.4  | 47.3  | 39.0  | 17%    |
| gpt-oss 25% <sup>†</sup> | 28.3        | 26.1    | 27.3  | 25.1  | 15.5  | 17%    |
| gpt-oss 40%              | 14.3        | 12.6    | 14.0  | 13.2  | 6.8   | 23%    |
| Qwen3 12.5% <sup>†</sup> | 166.2       | 146.3   | 134.1 | 109.1 | 100.4 | 30%    |
| Qwen3 25% <sup>†</sup>   | 84.2        | 74.8    | 74.1  | 62.8  | 44.3  | 24%    |
| Qwen3 43.75%             | 28.8        | 26.3    | 27.7  | 25.3  | 13.8  | 17%    |

Table 17. Partial foresight on the AIME routing the engine runs: host reads per token of admitting every miss and of MIN with one read, and the share of the gap between them that each window closes (Table 1). Recall 0.5: each future expert kept with probability 0.5 and replaced by a random one (fill) or left out (drop); means of three seeds. Linear: the ridge forecaster of Appendix I with a 4-token window.  $\kappa=1$ : the same exact 4-token window with the deployed margin instead of admitting every miss beyond it, as a share of its own gap.

| Budget      | Reads per token |       | Exact window $W$ (tokens) |      |      |      |      | Recall 0.5, fill / drop |             | Linear | $\kappa=1$ |
|-------------|-----------------|-------|---------------------------|------|------|------|------|-------------------------|-------------|--------|------------|
|             | every miss      | MIN   | 1                         | 2    | 4    | 8    | 16   | $W=8$                   | all         | $W=4$  | $W=4$      |
| gpt-oss 11% | 58.6            | 39.3  | 0.18                      | 0.31 | 0.52 | 0.81 | 0.98 | 0.29 / 0.29             | 0.43 / 0.48 | 0.13   | 0.62       |
| gpt-oss 25% | 28.2            | 16.0  | 0.07                      | 0.14 | 0.25 | 0.42 | 0.66 | 0.14 / 0.13             | 0.46 / 0.47 | 0.05   | 0.34       |
| Qwen3 12.5% | 150.0           | 100.3 | 0.31                      | 0.51 | 0.77 | 0.96 | 0.99 | 0.42 / 0.43             | 0.48 / 0.52 | 0.19   | 0.88       |
| Qwen3 25%   | 79.6            | 44.9  | 0.16                      | 0.28 | 0.46 | 0.70 | 0.93 | 0.29 / 0.28             | 0.49 / 0.53 | 0.11   | 0.60       |

**Speculation is not foresight.** Speculative decoding verifies  $K$  draft tokens in one forward pass, so a layer’s experts for  $K$  positions are requested together and a missed expert is read once for all of them. On the same traces, with drafts accepted at rate  $\alpha$  until the first rejection, batched verification saves an online cache a median 15–32% of its reads at  $K=8$  when rejected drafts cost nothing, the equivalent of a window of two to four tokens; once the rejected drafts’ experts are counted, the saving is 8–12% at  $\alpha=0.9$  and becomes a loss of 18–20% at  $\alpha=0.8$ , because the union of  $K$  positions’ expert sets is 20–79% of  $Kk$ . MIN itself gains little from batching (−7 to 15%). Systems that use drafts as a look-ahead sensor (Li et al., 2026) compare against a baseline that already pays for verification; ours charges the rejected drafts to a cache that does not speculate.

## I PARTIAL FORESIGHT IN DETAIL

**The window policies.** At step  $g$  the engine plans the step’s misses before it runs, from the trace’s record of step  $g$  (the routers give it layer by layer in a deployable system) and from a view of records  $g+1..g+W$  of the same problem. A miss is processed in request order: it takes a free slot; otherwise the victim is a resident not requested now that is not seen in the window (lowest decayed score), or, if every resident is seen, the one seen furthest; a miss seen in the window is admitted unless the victim is seen sooner, and a miss not seen is admitted if the victim is not seen either (admit every miss beyond the window; with the deployed margin  $\kappa=1$ , only if its score beats the victim’s by one). On the panel admissions are fetched in the step, so every miss is read once; on the deployed path of the follow-up the host’s fetch table splits the misses between the CPU and an in-step copy, and an admission served by the CPU is copied afterwards, a second read. In a degraded view each expert of a future record is kept with probability  $r$ , decided by a hash of (seed,  $g$ , record, layer, position), so step  $g$  sees one view throughout and step  $g+1$  an independent one; a dropped expert is replaced by a uniformly drawn one (fill: precision and recall  $r$ ) or left out (drop: recall  $r$ , precision 1). `scripts/value_map.py` replays the plan bit for bit: on a CPU build the engine’s reads equal the replay’s in 30 configurations at three budgets (`cputest099`), and on the panel’s GPUs, where the routing in the run can differ from the recorded trace by numerics, within 1.7% at both gpt-oss budgets. The panel ran the first 20 of the 30 problems; the table’s reads are on all 30.

**What the table says.** Horizon and accuracy trade: at gpt-oss 11% an 8-token window at recall 0.5 closes 0.29 of the gap, about what two exact tokens close (0.31), and the whole future at recall 0.5 closes 0.43. Fill and drop cost about the same for windows of up to 16 tokens; with the whole future in view at low recall, dropping costs less (the largest difference, 0.12,

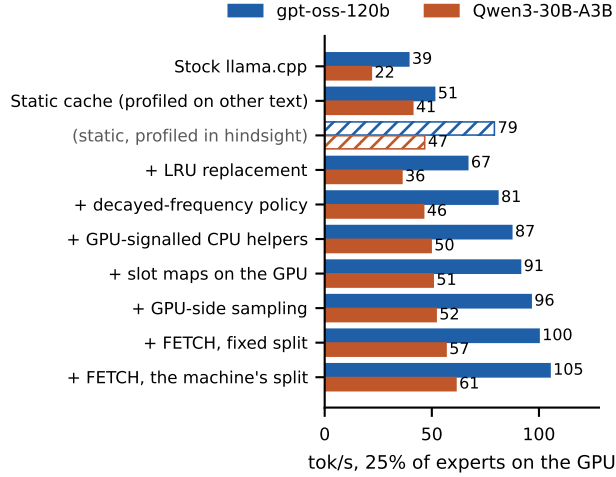


Figure 8. Ablation at 25% of experts on the GPU (15 AIME-25 problems, 256 tokens, host A). Each step adds one component to the one above; the hatched row is not a step: a static cache of each layer’s 32 most requested experts counted on the measured problems themselves.

is there). A larger cache needs a longer window: at 25% the same exact 4-token window closes 0.25 of the gap, against 0.52 at 11%.

**A realisable forecaster.** Ridge regression from the all-layer routing of step  $t$  to that of step  $t+h$  ( $h=1..8$ ), fitted on each model’s mixed-domain own text and applied to the AIME routing, forecasts the top  $k$  experts of each layer with a precision of 0.48 one token ahead and 0.26 four ahead on gpt-oss (0.54 and 0.32 on Qwen3); a decayed request count, the online policy’s own signal, scores 0.30 and 0.28. Driving the window policy, the ridge forecasts close 3–15% of the gap on gpt-oss and 6–19% on Qwen3 over windows of 1–8 tokens (`prereg/forecaster_linear.json`). Its precision falls with the horizon, so over a window it is closer to 0.3 than to its one-token value; at that average precision a window with random errors closes about as much on gpt-oss, and more on Qwen3, where the forecaster’s correct forecasts are mostly experts the cache already holds. A sequence model reports recall above 90% one token ahead with  $k+3$  candidates (Wang et al., 2026b); its precision is then at most  $k/(k+3)$ , 0.57 for gpt-oss and 0.73 for Qwen3. A recurrent forecaster in that spirit (a GRU of 512 units over the same all-layer vectors, one output head per horizon, trained on the same text; `scripts/forecaster_gru.py`) reaches a precision of 0.40 and 0.26 one and four tokens ahead on gpt-oss (0.46 and 0.31 on Qwen3), a recall at  $k+3$  of 0.53 and 0.54, and closes 2–11% and 6–17% of the gap: no better than the linear model. Our training text is smaller than SeqMoE’s and differs from the test text, so this bounds what such a model does across text, not what it can do in domain.

## J THE EXPERT CACHE IN MORE DETAIL

This section holds the measurements of Section 2 that the main text summarises: where the cache’s gain comes from, the choice of the FETCH split, the hosts with half the link, long outputs, FreeToken’s own headline model, two other llama.cpp-based systems, output parity, and the cost of prompts and batches. Job numbers refer to the scripts on the `gpu` branch.

**Where the gain comes from.** Figure 8 builds the system up from stock llama.cpp on one machine. Placing each layer’s hot experts is the first gain: a static cache of the 32 experts per layer most requested on other text (the models’ own MATH-500 outputs) runs  $1.31\times$  and  $1.89\times$  stock, which pins whole layers instead. Its hit rate (52% and 67%) depends on how well the profile matches the workload: profiled in hindsight on the measured problems it hits 75% and 72% and runs faster (hatched). Replacing experts online closes that gap without a profile, if admissions are rationed: LRU admits every miss, and on Qwen3 its copies raise host reads from 127 to 188 experts per token, so it runs  $0.88\times$  the static cache ( $1.30\times$  on gpt-oss); the decayed-frequency policy reads 41 and 118 per token, runs  $1.58\times$  and  $1.12\times$  the profiled static cache, and the hindsight

Table 18. The bound under each tightening, tokens per second on host B, and our cache as a percentage of the bound as published (exact optimum, highest probed host rate, datasheet GPU rate) and with every tightening at once. *no-evict*: Belady’s MIN with bypass that keeps a served expert for the step; *exact*: the same, free to evict an expert after serving it; *pool*: one pool of *LC* slots shared across layers; *median*:  $B_{\text{host}}$  the median of six concurrent probe samples instead of the highest; *GPU*: the GPU term from the measured all-in-VRAM time instead of the datasheet rate; *layers*: the per-layer sum, which no system without cross-layer prefetch can beat; *all*: exact, pool, median and GPU together (a warm start, not included, changes the bound by at most 1.4%).

| Model         | Experts on GPU | no-evict | exact | pool | median | GPU | layers | all | Ours, % of exact | % of all |
|---------------|----------------|----------|-------|------|--------|-----|--------|-----|------------------|----------|
| gpt-oss-120b  | 11%            | 169      | 172   | 182  | 159    | 172 | 163    | 168 | 41               | 41       |
| gpt-oss-120b  | 25%            | 427      | 430   | 474  | 397    | 279 | 365    | 278 | 25               | 39       |
| gpt-oss-120b  | 40%            | 519      | 519   | 519  | 517    | 279 | 500    | 278 | 29               | 55       |
| Qwen3-30B-A3B | 12.5%          | 92       | 96    | 101  | 89     | 96  | 93     | 93  | 42               | 43       |
| Qwen3-30B-A3B | 25%            | 209      | 214   | 234  | 197    | 192 | 193    | 191 | 30               | 33       |
| Qwen3-30B-A3B | 43.75%         | 308      | 308   | 308  | 307    | 192 | 298    | 191 | 35               | 57       |

cache runs 0.98 [0.95, 1.01] and 1.01 [0.98, 1.04] times as fast as it. On traces LRU misses only 2–11% more than decayed frequency at these budgets (Section 5); the rest of its cost here is the copies it admits. On host S, with everything else as in the full system, LRU runs 6–21% slower than decayed frequency across the six cells of Table 2 and trails FreeToken at four of them: the admission policy is most of the lead over FreeToken (77–84% of it at gpt-oss 25 and 40%, more than the whole of it at gpt-oss 11% and Qwen3 12.5 and 25%). The GPU-signalled CPU helpers, GPU slot maps, GPU sampling and FETCH add 2–9% each, and the model’s FETCH table 5–8% over the fixed one. The full system runs  $2.68\times$  (gpt-oss) and  $2.81\times$  (Qwen3) stock llama.cpp.

**How tight is the bound?** Table 18 changes one assumption at a time on host B. Pooled slots need 4.5–18.6% fewer reads and raise the host-bound limits by 5–10%; starting MIN warm changes it by at most 1.4%; the median probe sample instead of the highest lowers the host-bound limits by 8%. The GPU term moves most: at batch size 1 the all-in-VRAM runs reach only 52% (gpt-oss) and 61% (Qwen3) of the datasheet rate, and with the measured rate the limits at 25% and above fall to 279 and 192 tok/s, 8–9% above the all-in-VRAM speed. In the running example (gpt-oss at 25%) the limit on host B is 430 tok/s (279 with the measured GPU rate) and the cache runs 109 tok/s. With every expert in its slots the cache matches stock llama.cpp (0.997 $\times$  and 1.047 $\times$ ). The bound assumes an expert run on the CPU is read from DRAM (MIN’s shortest re-read distance is 0.5 GB of host traffic) and that computation runs at memory speed. The routing traces are each model’s own greedy text on the 30 prompts; replayed through a simulator of the deployed policy they reproduce the engine’s hit rates within 5.3 points.

**Reaching the read time.** The microbenchmark of Section 3 (`jobs/ec2/readsched.cu`) takes MIN’s misses per layer and step from `scripts/readsched.py` and runs no model. Usable cores minus two CPU threads read their share of host memory from an ordinary buffer, as the engine’s helpers do, and the copy engine copies the rest, 13.25 MB at a time, from pinned memory into GPU slots; both buffers rotate through 4 GiB so that no cache holds them. *Per layer* splits each layer’s experts, whole, between the two paths at the probe’s rates and waits for both before the next layer; *per token* sums the same split over the token and waits once; *link only* sends every read over PCIe, as admitting every miss does (MIN with one read sends only its admissions); *CPU only* reads everything on the CPU. Table 19 gives each host. Waiting per layer instead of per token costs at most 1 point of the host term on the two AMD hosts and 7 on the Intel host, whose link is the slowest; there, reading everything on the CPU beats the per-layer split. The split ignores which reads are admissions; Section 3 gives the link-aware bound, which sends every admission over the link (`scripts/linkaware.py`, `prereg/linkaware.json`).

**Choosing the split per machine.** On host B, the model’s table beats the fixed table we used before at every budget of both models, by 3.3–8.0% (launch 1). Table 20 shows why at one budget: a single missed expert should run on the CPU (about 155  $\mu$ s, hidden behind the GPU’s other experts) rather than be copied (about 225  $\mu$ s, on the critical path); that entry alone

Table 19. Can the bound’s read time be reached (job 102)? MIN’s per-layer host reads of gpt-oss (first 4,000 steps of the AIME routing), replayed without the model through CPU threads and the copy engine. Each mode’s share of the bound’s host term ( $R^*S/B_{\text{host}}$ , ms per token in the column *Term*); *Model*: the per-layer mode predicted from the probe’s rates. Two repetitions of each mode agree within 0.9%.

| Host     | Link/<br>CPU | $C$ | Term<br>(ms) | Split<br>per layer | Split<br>per token | Link<br>only | CPU<br>only | Model |
|----------|--------------|-----|--------------|--------------------|--------------------|--------------|-------------|-------|
| Pd again | 0.40         | 14  | 7.04         | 86%                | 93%                | 35%          | 92%         | 98%   |
| Pd again | 0.40         | 32  | 2.84         | 89%                | 93%                | 35%          | 92%         | 98%   |
| 9950X    | 1.02         | 14  | 9.70         | 96%                | 97%                | 95%          | 86%         | 94%   |
| 9950X    | 1.02         | 32  | 3.91         | 96%                | 97%                | 94%          | 86%         | 92%   |
| 9800X3D  | 0.55         | 14  | 9.56         | 95%                | 95%                | 53%          | 91%         | 95%   |
| 9800X3D  | 0.55         | 32  | 3.85         | 93%                | 94%                | 53%          | 91%         | 94%   |

Table 20. FETCH tables at 43.75% on Qwen3-30B-A3B (launch 1): entry  $n$  is how many of a layer’s  $n$  missed experts are copied over PCIe; the rest run on the CPU. Ratio to the fixed table, paired, 95% CI.

| Table                      | $f(0..8)$                 | tok/s | vs fixed             |
|----------------------------|---------------------------|-------|----------------------|
| Lite                       | 0, 0, 1, 1, 1, 2, 2, 3, 3 | 109.0 | 1.057 [1.052, 1.062] |
| Model                      | 0, 0, 1, 1, 2, 3, 3, 4, 5 | 108.0 | 1.048 [1.041, 1.055] |
| Model, fetch a single miss | 0, 1, 1, 1, 2, 3, 3, 4, 5 | 104.1 | 1.010 [1.007, 1.012] |
| Fixed                      | 0, 1, 1, 2, 3, 3, 4, 5, 6 | 103.1 | 1                    |
| None (CPU only)            | 0, . . . , 0              | 100.7 | 0.977 [0.970, 0.984] |
| All (fetch every miss)     | 0, 1, . . . , 8           | 98.7  | 0.957 [0.953, 0.962] |

is worth 3.7%. Fetching everything, FreeToken’s offload strategy expressed in our engine, is the worst table, and running everything on the CPU the second worst. The model’s table was within 1% of the best of five alternatives on launch 1 and tied with it on launch 2. Given each run’s measured misses, Eq. (5) ranks the tables correctly except law against lite, which are within 1%, and overstates their differences about  $1.8\times$ .

**Machines with half the link.** On RTX 5090s whose links we measured at 27–30 GB/s instead of 53 (jobs 085, 085b and 088: gpt-oss next to a Core i9-14900K, Qwen3 next to a Ryzen 9 7900 in a PCIe 4.0 slot), the model’s tables fetch less (gpt-oss 0, 0, 0, 1, 1, Qwen3 0, 0, 1, 1, 2, 2, 3, 3, 4) and beat the fixed table by 21–34% on gpt-oss and 9–18% on Qwen3, about twice the gain on host B for Qwen3 and four to six times for gpt-oss: the per-machine split matters most where the link is slow. On the Ryzen, ours leads FreeToken by  $1.04\text{--}1.08\times$ ; with the fixed table it would trail at 12.5%. On the Core i9 the first comparison (job 085) was confounded: FreeToken’s hybrid backend ran gpt-oss at 12–66 tok/s, below its own offload backend (25–86), because its CPU executor spread 23 threads across the 8 performance and 16 efficiency cores and its calibration then sent two thirds of the misses over the slow link, the same failure Appendix C reports for llama.cpp; the  $1.4\text{--}2.0\times$  lead that gave us measured a thread configuration. We reran it (job 088, a second rental of the same CPU model) with FreeToken’s executor on 8 threads pinned to the performance cores: its hybrid backend then runs  $1.4\text{--}3.1\times$  faster than before (37.5, 62.1 and 92.5 tok/s; its calibration now copies 51.7% of misses), beating its own offload backend at 11 and 25% and tying it at 40%, and ours leads its better configuration by  $1.58, 1.52$  and  $1.51\times$  (lowest interval bound 1.47). Restricted to the CPU alone, FreeToken’s 8-thread executor decodes 13.7 tok/s at every budget. llama.cpp on the same rental runs 40.0 tok/s with one thread per performance core ( $-9.4\%$  from the model’s prediction) and 32.8 with all 24 threads ( $+25.9\%$ ): the hybrid-core failure is a thread-count choice, and the model holds once the choice is right. The law predicted

llama.cpp on the Ryzen within +2.7%. Figure 5 puts every same-machine comparison on one axis.

**Long outputs.** Our lead does not come from the first 256 tokens. On gpt-oss, on ten held-out hard problems (AIME 2022, problems 11–15 of both exams) that neither system has finished reasoning about after 2,048 tokens (at most 1 of 10 reach an answer), ours runs 1.36 [1.34, 1.38], 1.32 [1.30, 1.34] and 1.19 [1.17, 1.21] times FreeToken at 11, 25 and 40% over all 2,048 tokens, and 1.37 [1.34, 1.39], 1.33 [1.31, 1.35] and 1.19 [1.17, 1.22] over tokens 257–2,048 (job 083, host A); both engines stream one event per token. On Qwen3, over tokens 257–2,048 of ten held-out MATH-500 problems, ours runs 1.043 [1.020, 1.070], 1.056 [1.036, 1.077] and 1.054 [1.025, 1.086] times FreeToken at 12.5, 25 and 43.75%. Both leads are larger than over the same runs’ first 256 tokens. Turning off FreeToken’s prefill overlap, which lends it cache slots during each prompt, changes its overall speed by 1% or less.

**FreeToken’s own headline model.** On FreeToken’s own headline model, Qwen3.6-35B-A3B in BF16 (256 experts of 6.3 MB per layer, hybrid linear attention), the two systems are close: ours runs 1.030 [1.023, 1.038] times FreeToken at 12.5% and ties at 25 and 37.5% (1.010 [0.998, 1.021], 1.003 [0.994, 1.013]), at 1.98–2.48× llama.cpp (job 086, host A). FreeToken as shipped, with its automatic backend and cache size, runs this model at 93.8 tok/s on that machine, above the 77–83 tok/s its paper reports on an RTX 5090, so our setup does not handicap it; FreeToken’s fetch cap and thread count were left at their defaults, while our FETCH table was swept. Our lead is therefore model-dependent: largest with few large experts (gpt-oss, top-4 of 128), smallest with many small ones.

**FreeToken tuned.** On an RTX 5090 with a Ryzen 9 9950X (job 098, one launch, configurations in a shuffled order per cell) FreeToken ran in five settings per cell: hybrid with its calibrated fetch split (its default), hybrid with the fetch cap fixed at one and at two, hybrid with 8 CPU threads instead of one per physical core, and offload. Its best setting was the calibrated hybrid with 8 threads at three cells and offload at the other three (gpt-oss 25 and 40%, Qwen3 43.75%), at most 1.04× its Table 2 setting; the fixed caps were never better than its own split by more than 0.3%. Ours over FreeToken’s best: 1.17, 1.17 and 1.09 on gpt-oss, 1.02, 1.04 and 0.98 on Qwen3, within 0.04 of host S’s ratios at every cell. With every expert on the CPU, stock llama.cpp decodes gpt-oss 1.13× and Qwen3 0.99× as fast as FreeToken’s CPU backend.

**Other systems and outputs.** Two newer llama.cpp-based systems, Pipelined Sharding (Ukarande et al., 2026) and leloch’s expert cache (leloch, 2026a), run gpt-oss-120b at 1.28–1.45× stock llama.cpp; ours ran 1.42–2.34× the better of them (job 077). KTransformers (Chen et al., 2025a) (kt-kernel 0.7.0, exact BF16 CPU experts, the same number of experts statically on the GPU) ran Qwen3 in BF16 at 0.98–1.01× stock llama.cpp on an RTX 5090 with a Ryzen 9 9950X, and ours at 2.2–4.4× it (job 078); both static placements read the same CPU expert bytes from the same DRAM. Its published speeds rely on AMX Xeons and INT4/INT8 CPU weights, which that host lacks and which change the model, so this is not its best setting. Teacher-forced on fixed text (1536 steps per model) against stock llama.cpp with every weight in the VRAM of an RTX PRO 6000 (job 090), our cache at 25% has a mean KL divergence over the full vocabulary of 0.0019 nats on gpt-oss and 0.0005 on Qwen3 (99.9th percentile 0.121 and 0.059, maximum 0.21 and 0.13), picks the same next token on 98.5% and 99.3% of steps, and changes the loss by +0.16% and +0.19%. Stock llama.cpp’s own --n-cpu-moe placement diverges from the same reference by as much (mean KL 0.0017 and 0.0008, agreement 98.8% and 99.3%): the differences are those of running experts on the CPU (the helpers quantise activations to Q8\_0 where the GPU path uses Q8\_1), not of the cache. Task accuracy was not measured.

**Prompts and batches.** The cache serves single-token steps; prompts and batches above one run llama.cpp’s stock graph with every expert in host memory, and pay for the cache’s resident experts and its one-thread CPU side. In llama-batched-bench at 25% on host S (job 089, both models) our prefill at 512 and 2,048 tokens runs 79–83% of stock llama.cpp’s and our decode at 2 and 4 parallel sequences 77–79% of it (the tool feeds both engines the same random tokens).

## K THE AUDIT’S ROWS

Table 21 lists the 52 published measurements of Section 3 with their bounds. The sources are Fiddler (Kamahori et al., 2024), ProMoE (Song et al., 2024a), KTransformers (Chen et al., 2025a), HybriMoE (Zhong et al., 2025), Fate (Fang et al., 2025), SP-MoE (Chen et al., 2025b), MoE-SpeQ (Wang et al., 2025), the CPU-GPU collaborative inference of Huang et al. (2025), FreeToken (Yang et al., 2026), SeqMoE (Wang et al., 2026b), NVIDIA’s pipelined sharding (Ukarande et al., 2026), a llama.cpp fork (leloch, 2026a) and a llama.cpp pull request (csantiago078, 2026). The in-class subset of Section 3 (20 rows)

## Where the Seconds Go

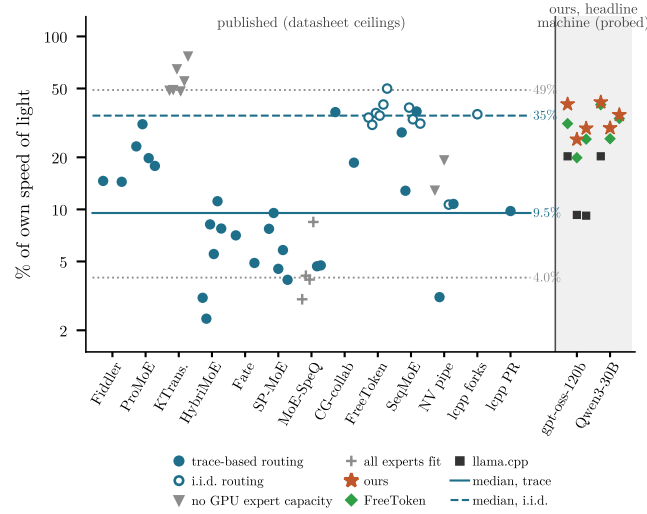


Figure 9. Published MoE-offloading measurements (52 rows from 13 sources) as a percentage of their own bound, computed from the paper’s hardware at datasheet rates and, where a routing trace of the model exists, the per-layer Belady optimum (filled); hollow points assume uniform routing, which overstates the optimum’s reads and so the percentage; the dotted lines are the class medians (no GPU expert capacity, 49.0%; every expert resident, 4.0%). Right: our cache, FreeToken and stock llama.cpp on host B against the probed per-layer limit of Table 2 (the pooled bound FreeToken’s and llama.cpp’s designs allow is 0–10% higher).

is the trace-scored rows less the lossy (Fate) and speculative (SP-MoE, MoE-SpeQ) ones; over all 29 trace-scored rows the median is 9.5%. These systems mostly read on demand, so Eq. (2) would place them higher. Three caveats apply to every row: the ceilings are datasheet ones where ours are probed; the trace is the model’s own sampled text, sometimes of a sibling variant (an FP8 trace for a Q4 deployment); and systems whose cache is pooled across layers, whose placement pins whole layers, or which verify several tokens per pass are scored against the per-layer one-token bound, which is not theirs.

Table 21: The audit’s rows (Section 3): every published batch-1 measurement we scored, with the system’s stated hardware, its experts per layer resident on the GPU ( $C$ ), the model’s experts and top- $k$ , the published speed, the bound at datasheet-physical ceilings (GPU datasheet bandwidth, the top of the host-DRAM band, datasheet PCIe, no latencies; `scripts/audit.py`), how the optimum’s reads were obtained (*trace*: Belady’s MIN with bypass on an on-policy routing trace of the model; *uniform*: independent uniform routing, which overstates the reads and so the percentage; *no slots*: no GPU expert capacity, the CPU-only rate is the bound; *all fit*: every expert resident, nothing offloaded) and the percentage of the bound. Medians: trace 9.5% ( $n=29$ ), uniform 34.8% (11), no slots 49.0% (8), all fit 4.0% (4), all rows 14.5% (52). Rows below 5% are more likely configuration mismatches with the bound’s assumptions (precision, dense layers on the CPU, end-to-end timing) than inefficiency, and are not adjudicated.

| System                             | Model                 | GPU             | $C$ | $E/k$ | tok/s | bound | reads    | %    |
|------------------------------------|-----------------------|-----------------|-----|-------|-------|-------|----------|------|
| Fiddler (Kamahori et al., 2024)    | Mixtral-8x7B          | Quadro RTX 6000 | 1   | 8/2   | 2.5   | 17    | trace    | 14.6 |
|                                    | Mixtral-8x7B          | RTX 6000 Ada    | 3   | 8/2   | 8.3   | 57    | trace    | 14.4 |
| ProMoE (Song et al., 2024a)        | DeepSeek-V2-Lite      | RTX 4090        | 45  | 64/6  | 49.9  | 216   | trace    | 23.1 |
|                                    | Qwen1.5-MoE-A2.7B     | RTX 4090        | 46  | 60/4  | 67.2  | 216   | trace    | 31.1 |
|                                    | Qwen2-57B-A14B        | RTX 4090        | 42  | 64/8  | 26.9  | 136   | trace    | 19.8 |
|                                    | Mixtral-8x7B          | RTX 4090        | 6   | 8/2   | 27.0  | 152   | trace    | 17.8 |
| KTransformers (Chen et al., 2025a) | DeepSeek-V3-0324      | A100 40GB       | 0   | 256/8 | 5.9   | 12    | no slots | 48.8 |
|                                    | DeepSeek-V2.5-1210    | A100 40GB       | 0   | 160/6 | 11.9  | 24    | no slots | 49.2 |
|                                    | Qwen2-57B-A14B        | A100 40GB       | 0   | 64/8  | 22.9  | 35    | no slots | 64.9 |
|                                    | DeepSeek-V3-0324      | RTX 4080        | 0   | 256/8 | 16.1  | 33    | no slots | 48.3 |
|                                    | DeepSeek-V2.5-1210    | RTX 4080        | 0   | 160/6 | 17.6  | 32    | no slots | 55.3 |
|                                    | Qwen2-57B-A14B        | RTX 4080        | 0   | 64/8  | 37.0  | 48    | no slots | 76.9 |
| HybriMoE (Zhong et al., 2025)      | DeepSeek-V2-Lite-Chat | RTX A6000       | 16  | 64/6  | 14.5  | 470   | trace    | 3.1  |

continued

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| System                                     | Model                           | GPU                    | $C$ | $E/k$  | tok/s | bound | reads    | %    |
|--------------------------------------------|---------------------------------|------------------------|-----|--------|-------|-------|----------|------|
|                                            | DeepSeek-V2-Lite-Chat           | RTX A6000              | 48  | 64/6   | 15.3  | 655   | trace    | 2.3  |
|                                            | Mixtral-8x7B-Instruct           | RTX A6000              | 2   | 8/2    | 4.4   | 53    | trace    | 8.2  |
|                                            | Mixtral-8x7B-Instruct           | RTX A6000              | 6   | 8/2    | 7.3   | 133   | trace    | 5.5  |
|                                            | Qwen2-57B-A14B-Instruct         | RTX A6000              | 16  | 64/8   | 9.6   | 86    | trace    | 11.1 |
|                                            | Qwen2-57B-A14B-Instruct         | RTX A6000              | 48  | 64/8   | 9.0   | 116   | trace    | 7.7  |
| Fate (Fang et al., 2025)                   | Qwen1.5-MoE-A2.7B               | RTX 3090               | 46  | 60/4   | 14.0  | 198   | trace    | 7.7  |
|                                            | Qwen1.5-MoE-A2.7B               | GTX 1080 Ti            | 15  | 60/4   | 3.7   | 76    | trace    | 4.9  |
| SP-MoE (Chen et al., 2025b)                | Mixtral-8x7B                    | RTX 3090               | 1   | 8/2    | 1.2   | 16    | trace    | 7.7  |
|                                            | Mixtral-8x7B                    | RTX 4090               | 1   | 8/2    | 1.3   | 13    | trace    | 9.5  |
|                                            | Mixtral-8x7B                    | A100 40GB              | 3   | 8/2    | 1.5   | 32    | trace    | 4.5  |
|                                            | Phi-3.5-MoE                     | RTX 4090               | 3   | 16/2   | 3.3   | 57    | trace    | 5.8  |
|                                            | DeepSeek-V2-Lite                | A100 40GB              | 7   | 64/6   | 5.6   | 142   | trace    | 3.9  |
| MoE-SpeQ (Wang et al., 2025)               | DeepSeek-V2-Lite                | A100 40GB              | 64  | 64/6   | 10.1  | 335   | all fit  | 3.0  |
|                                            | DeepSeek-V2-Lite                | A100 40GB              | 64  | 64/6   | 13.8  | 335   | all fit  | 4.1  |
|                                            | Qwen1.5-MoE-A2.7B               | A100 40GB              | 60  | 60/4   | 13.2  | 335   | all fit  | 3.9  |
|                                            | Qwen1.5-MoE-A2.7B               | A100 40GB              | 60  | 60/4   | 28.3  | 335   | all fit  | 8.5  |
|                                            | Phi-3.5-MoE                     | A100 40GB              | 7   | 16/2   | 6.0   | 129   | trace    | 4.7  |
|                                            | Phi-3.5-MoE                     | A100 40GB              | 7   | 16/2   | 6.1   | 129   | trace    | 4.7  |
| CPU-GPU collab. (Huang et al., 2025)       | Mixtral-8x7B                    | RTX 4090               | 1   | 8/2    | 4.8   | 13    | trace    | 36.5 |
|                                            | Phi-3.5-MoE                     | RTX 4090               | 3   | 16/2   | 10.4  | 56    | trace    | 18.6 |
| FreeToken (Yang et al., 2026)              | Qwen3.6-35B-A3B                 | RTX 5090               | 94  | 256/8  | 77.1  | 227   | uniform  | 34.0 |
|                                            | DeepSeek-V4-Flash               | RTX 5090               | 28  | 256/6  | 24.9  | 81    | uniform  | 30.8 |
|                                            | Qwen3.6-35B-A3B                 | RTX 5090               | 104 | 256/8  | 73.8  | 205   | uniform  | 36.1 |
|                                            | Qwen3.6-35B-A3B                 | RTX 4090               | 72  | 256/8  | 42.9  | 123   | uniform  | 34.8 |
|                                            | Qwen3.6-35B-A3B                 | RTX 4060 Laptop        | 60  | 256/8  | 39.3  | 97    | uniform  | 40.4 |
|                                            | GLM-5.2                         | RTX PRO 6000 Blackwell | 36  | 256/8  | 14.9  | 30    | uniform  | 49.9 |
| SeqMoE (Wang et al., 2026b)                | Qwen3-30B-A3B-FP8               | RTX 4090               | 57  | 128/8  | 104.1 | 374   | trace    | 27.9 |
|                                            | Qwen3-30B-A3B-FP8               | RTX 4090               | 19  | 128/8  | 47.9  | 374   | trace    | 12.8 |
|                                            | Qwen3.6-35B-A3B-BF16            | RTX 5090               | 102 | 256/8  | 118.3 | 305   | uniform  | 38.8 |
|                                            | Qwen3.6-35B-A3B-BF16            | RTX 5090               | 38  | 256/8  | 70.2  | 212   | uniform  | 33.2 |
|                                            | GPT-OSS-120B-MXFP4              | RTX PRO 6000 Blackwell | 57  | 128/4  | 139.2 | 378   | trace    | 36.8 |
|                                            | DeepSeek-V4-Flash               | RTX PRO 6000 Blackwell | 115 | 256/6  | 57.4  | 183   | uniform  | 31.3 |
| Pipelined sharding (Ukarande et al., 2026) | Qwen3-30B-A3B-Instruct-2507     | RTX 5090               | 0   | 128/8  | 25.7  | 199   | no slots | 12.9 |
|                                            | Qwen3-30B-A3B-Instruct-2507     | RTX 5090               | 44  | 128/8  | 32.1  | 1032  | trace    | 3.1  |
|                                            | Qwen3-235B-A22B-Instruct-2507   | RTX 5090               | 0   | 128/8  | 7.7   | 40    | no slots | 19.3 |
|                                            | Qwen3-235B-A22B-Instruct-2507   | RTX 5090               | 42  | 128/8  | 11.5  | 108   | uniform  | 10.7 |
|                                            | Qwen3-30B-A3B-Instruct-2507     | RTX 5070 Ti            | 107 | 128/8  | 54.9  | 511   | trace    | 10.8 |
| llama.cpp fork (leloch) (leloch, 2026a)    | Laguna S 2.1                    | RTX 5090               | 66  | 256/10 | 31.3  | 88    | uniform  | 35.5 |
| llama.cpp PR 27861 (csantiago78, 2026)     | Qwen3-30B-A3B Q4_K_M derivative | Radeon RX 7600 8GB     | 32  | 128/8  | 16.5  | 169   | trace    | 9.8  |

## L ARTIFACT DESCRIPTION

- **System.** A patch to llama.cpp commit 4da6337 (the expert cache, FETCH, the mailbox and the llama-ec-bench instrumentation: teacher-forced loss, logit dumps, routing lookahead records); built with CUDA 13.0 for compute capability 12.0.
- **Jobs.** One shell script per job; each installs FreeToken at commit 0d652e7, downloads the models from Hugging Face, runs the bandwidth probes, writes any on-machine prediction, runs the client, and stores raw rows (one JSON line per problem), server logs and engine counters. A small runner rents a Vast instance, streams a heartbeat, fetches the results and only then destroys the instance.
- **Analysis.** scripts/\*\_stats.py compute every interval from the raw rows; scripts/perlayer\_model.py, speed\_limit\_v2.py, shapley\_gap.py, policy\_study.py, foresight\_exact.py, batchk\_trace.py, fig\_audit\_sol.py and scorecard.py produce the model comparison, the limit, the accounting, the trace studies, the audit figure and the scorecard; scripts/wsg\_tables.py and wsg\_numbers2.py generate every table, figure and number in this paper from their outputs. The audit’s 52 published rows and the physical speed-of-light model they are scored against are data/published\_measurements.json, data/audit/ and scripts/audit.py.
- **Cost.** The 88 rentals (53 distinct machines) of jobs 058–101 cost 70.8 US dollars at their hourly GPU prices, by the

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ledger in the repository; storage and download charges, which the ledger omits, add to this.

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